Savitribai Phule Pune University, Pune

Faculty of Commerce and Management

Bachelor of Business Administration in Computer Application

(BBA - CA)

Revised Curriculum (2024 Pattern as per NEP-2020)

w.e.f. Academic Year: 2024-2025

Preamble:

In the rapidly evolving landscape of the digital age, the BBA program is meticulously designed to bridge the gap between commerce and technology. This program aims to equip students with a comprehensive understanding of both domains, fostering a unique blend of skills that are highly sought after in today's competitive job market. The program focuses on providing students with in-depth knowledge of computer applications, programming, and software development while integrating essential commerce subjects such as business mathematics, financial accounting, and management principles. This interdisciplinary approach ensures that graduates are not only proficient in technical skills but also possess a solid foundation in business operations and management. A key feature of this program is its emphasis on practical and project-based learning. Students engage in hands-on lab work, field projects, and internships that allow them to apply theoretical knowledge to real-world scenarios. This experiential learning approach prepares students to tackle complex business problems with innovative technological solutions. Moreover, the program fosters the development of essential soft skills such as effective communication, teamwork, leadership, and ethical decision-making. These skills are critical for personal and professional growth, enabling graduates to navigate the dynamic and often challenging business environments with confidence and integrity. The BBA - Computer Application program also promotes lifelong learning and adaptability, encouraging students to stay abreast of technological advancements and industry trends. By instilling a habit of continuous learning, the program prepares students to adapt to new tools, technologies, and methodologies throughout their careers. In essence, the BBA - Computer Application program aims to create well-rounded professionals who are equipped with the technical expertise, business acumen, and soft skills necessary to excel in the ever-changing landscape of commerce and technology. Graduates of this program will be poised to contribute effectively to their organizations and society, driving innovation and growth in their respective fields.

Following aspects highlight the importance of commercial education:

1. Academic Rigor and Excellence: Commercial education provides a rigorous academic curriculum that equips students with a comprehensive understanding of business theories, principles, and practices. Through innovative teaching methods and experiential learning opportunities students excel in dynamic and competitive global business environments.

2. Ethical Leadership and Social Responsibility: Students are instilled in the importance of ethical decision-making, integrity, and corporate social responsibility. Our program emphasizes the significance of ethical leadership and the impact of business practices on society and the environment.

3. Critical Thinking and Problem-Solving Skills: We foster the development of critical thinking, analytical reasoning, and problem-solving skills essential for effective decision-making in complex

business situations. Students learn to evaluate information, analyze data, and formulate strategic solutions to real-world challenges.

4. Global Perspective and Cultural Awareness: Recognizing the interconnectedness of the global economy, we emphasize the development of a global mindset and cultural competence among the students. Our curriculum integrates international business concepts and opportunities for cross-cultural learning experiences.

5. Professional Development and Career Readiness: Through internships, professional development workshops, and networking opportunities, students are facilitated the acquisition of practical skills and industry-specific knowledge necessary for professional growth and advancement.

6. Innovation and Entrepreneurship: Encouraging creativity and innovation, we inspire entrepreneurial thinking and the ability to identify and seize opportunities in the marketplace. Our program supports aspiring entrepreneurs in developing business plans and launching ventures that contribute to economic growth and innovation.

7. Continuous Learning and Adaptation: Committed to continuous improvement and adaptation to meet the evolving demands of the business world. Our faculty engage in scholarly research and professional development to ensure that our curriculum remains relevant and responsive to industry trends and technological advancements.

8. Constant Learning: Commerce is a field that requires continuous learning and adaptation to stay competitive. Business education instills a mindset of lifelong learning, encouraging individuals to stay updated about industry trends, new technologies, and evolving business practices.

Objectives of the Programme:

1. To equip students with comprehensive knowledge in computer applications, including programming languages such as C, C++, Java, and Python.
2. To provide hands-on experience with database management systems, web development tools, and software engineering.
3. To foster the ability to solve complex problems using structured programming and algorithmic approaches.
4. To enable students to analyze and develop efficient solutions in business and IT environments.
5. To blend core commerce subjects like business mathematics, financial accounting, and management principles with IT skills.
6. To prepare students to leverage technology in managing and analyzing business operations.
7. To incorporate practical sessions and lab work to apply theoretical concepts in real-world scenarios.
8. To encourage project-based learning through field projects and internships, focusing on web applications, mobile app development, and digital marketing.
9. To improve business communication skills through courses designed to enhance written and verbal communication.
10. To instill an understanding of business ethics, environmental awareness, and gender sensitization.
11. To offer a variety of elective courses and open electives to allow students to explore interdisciplinary areas.
12. To provide exposure to vocational skills, such as office automation tools, web technology, and .NET programming.
13. To lay a strong foundation for students aiming to pursue higher studies in commerce and computer applications.

14. To equip students with the necessary skills to excel in professional careers in IT, software development, data analysis, and business management.
15. To instill a habit of continuous learning to keep pace with technological advancements and evolving industry standards.
16. To prepare students to adapt to new tools, technologies, and methodologies in the field of commerce and IT.

Program Outcomes:

1. Graduates will demonstrate a thorough understanding and ability to apply core concepts in programming languages, database management systems, and software development.
2. Graduates will be able to analyze business problems, develop efficient algorithms, and implement solutions using appropriate programming techniques.
3. Graduates will effectively combine principles of commerce with modern IT practices to enhance business processes and decision-making.
4. Graduates will have practical experience in handling projects related to web development, mobile applications, and digital marketing, with a capability to manage and execute projects efficiently.
5. Graduates will possess strong written and verbal communication skills, essential for professional business environments, including report writing, presentations, and interpersonal communication.
6. Graduates will understand and adhere to ethical practices in business and IT, with a keen awareness of environmental issues and gender sensitivity.
7. Graduates will have exposure to a range of subjects and elective courses, providing a broad perspective and the ability to approach problems from various disciplinary angles.
8. Graduates will be prepared for employment in IT, software development, data analysis, business management, and related fields, with the skills to thrive in a professional environment.
9. Graduates will be committed to lifelong learning, staying current with technological advancements and adapting to new tools and methodologies.
10. Graduates will have the ability to work effectively in teams, exhibiting leadership skills and contributing to collaborative projects.
11. Graduates will be adept at applying theoretical knowledge in practical settings, utilizing hands-on experience gained through lab work, projects, and internships.
12. Graduates will exhibit critical thinking skills and a creative approach to problem-solving, fostering innovation in their work.
13. Graduates will have a solid foundation in business operations, financial accounting, and management principles, enabling them to contribute to organizational success.

Introduction

The BBA - CA Degree Program (2024 Pattern) will be introduced in the following order:

Sr. No.	BBA - CA Degree Program	Academic Year
A	First Year BBA - CA	2024-2025
B	Second Year BBA - CA	2025-2026
C	Third Year BBA - CA	2026-2027
D	Fourth Year BBA - CA	2027-2028

Eligibility

a) No Candidates shall be admitted to the First Year of the BBA - CA Degree Program (2024 Pattern) unless he / she has passed the Higher Secondary School Certificate Examination of the Maharashtra State Board or equivalent or University with English as a passing Course.
b) No candidate shall be admitted to the Third Semester Examination of the Second Year unless he / she has cleared First Two Semesters satisfactorily for the course at the college affiliated to this University.
c) No student shall be admitted to the Third Year BBA – CA (Fifth Semester) Degree Program (2024 Pattern) unless he / she has cleared all the papers of First and Second Semester Examination of F.Y. BBA - CA
d) No candidate shall be admitted to the Fifth Semester Examination of the Third Year unless he / she has cleared the first Two Semesters satisfactorily of Second Year for the Program at the college affiliated to this University.
e) No candidate shall be admitted to the Fourth Year BBA - CA (Seventh Semester) Degree Program (2024 pattern) unless he / she has cleared all the papers of Third and Fourth Semester Examination of S.Y. BBA- CA

Teaching Methodology

The Teacher can use the following Methods as Teaching Methodology:

- Class Room Lectures
- Demonstration for programming course
- Guest Lectures of Professionals, Industry Experts etc.
- Teaching with the help of ICT tools
- Visits to various Professionals Units, Companies and Business / Industry Units
- Group Discussion / Debates
- Assignments, Tutorials, Presentations, Role Play etc.
- YouTube Lectures developed by MHRD, UGC, Government of Maharashtra, University etc.
- Analysis of Case Studies

Examination

1) A student cannot appear for the Semester End Examination unless he / she has maintained at least 75% attendance during the teaching period of that course. If a student fails to maintain attendance up to 75%, at the time of filling of Examination Forms, an undertaking from the student should be taken stating that he / she will be allowed to appear for Examination subject to fulfillment of required attendance criteria during the remaining period of teaching of the course.
2) Each credit will be evaluated for 25 Marks.
3) Each course will have a distribution of 30:70 for CIE and SEE.
4) To pass a course, the student must obtain at least 40% Percent marks in the CIE and SEE separately.
5) If a student misses CIE examination, he / she will have a Second Chance with the permission of the teacher concerned only. Such a Second Chance shall not be the right of the student; it will be the discretion of the teacher concerned only rather than the Head of the Department or Principal to give or not to give Second Chance to a student to appear for Internal Assessment.
6) A student cannot register for the Third, Fifth and Seventh Semester, if he / she fails to complete 50% credits of the total credits expected to be ordinarily completed within Two Semesters.
7) No student shall be admitted to the Fifth Semester Examination of the Third Year unless he / she has cleared First Two Semesters.

8) No student shall be admitted to the Fourth Year BBA - CA (Seventh Semester) Degree Program (2024 Pattern) unless he / she has cleared all the papers of Third and Fourth Semester Examination of S.Y. BBA - CA and has satisfactorily kept terms for the Third Year (Fifth and Sixth Semester).
9) There shall be revaluation of the Answer Scripts of Semester-End Examination but not of Answer Scripts of Internal Assessment Papers as per Ordinance No. 134 A and B.

A.T.K.T. Rules

The present relevant ordinances issued by the SPPU pertaining to ATKT are applicable.

University Terms:

The dates for the commencement and conclusion of the First and the Second Terms shall be as determined by the University Authorities. Only duly admitted students can keep to the terms. The present relevant ordinances pertaining to the grant of terms will be applicable.

Verification and Revaluation

The candidate may apply for verification and revaluation or result through Principal of the College which will be done by the University as per ordinance framed in that behalf.

Restructuring of Courses

This revised course structure shall be made applicable to the colleges implementing 'Restructured Programme at the Undergraduate Level from June 2024. The Colleges under the Restructured Programme which have revised their structure in the light of the "2024 Pattern" shall be introduced with effect from Academic Year 2024-25.

Standard of Passing

- A candidate is required to obtain 40% Marks in Internal Assessment, Practical Examination and Semester End University Examination.
- It means that passing separately at Internal Assessment, Practical Examination and Semester End University Examination is compulsory.

Methods of Evaluation, Passing, and Evaluation Criteria

The evaluation of students will be done on Three Times during each Semester:

- Internal Assessment (Internal)
- Practical Examination (If applicable)
- Semester End University Examination (External)

For Semester End University Examination, question papers will be set for Seventy Percent of the Total Marks allotted for the course.

Evaluation will be done on a continuous basis Three Times during each Semester. Internal Assessment will be of Thirty Percent of the Total Marks allotted for the subject. The colleges need to adopt any Two Methods out of the following Methods for Internal Assessment:

- Offline Written Examination
- Power Point Presentations
- Assignments / Tutorials
- Oral Examination
- Open Book Test
- Offline MCQ Test
- Group Discussion
- Analysis of Case Studies

Programme Structure

FYBBA-CA Semester I							
Course Type	Course	Paper Title	Hours / Week	Credits	Internal	External	Total
Major Mandatory (06)	Major Mandatory 1	Problem solving using C	3	2	15	35	50
	Major Mandatory 2	Data Base Management System	3	2	15	35	50
	Major Mandatory 3 (Practical)	Computer Laboratory based on C Programming and Data Base Management System (DBMS)	5	2	15	35	50
Open Elective (OE)	Open Elective 1	Business Mathematics	3	2	15	35	50
	Open Elective 2	Business Statistics	3	2	15	35	50
Vocational Skill Development Course (VSC)	Vocational Skill Development Course	Office Automation tools	5	2	15	35	50
Skill Enhancement Course (SEC)	Skill Enhancement Course (SEC)	Programming Principles and algorithm	3	2	15	35	50
Ability Enhancement Course (AEC)	Ability Enhancement Course (AEC)	Business Communication Skills-I	3	2	15	35	50
Value Education Course (VEC)	Value Education Course (VEC)	Environmental Awareness	3	2	50	0	50
Indian Knowledge System (IKS)	Indian Knowledge System (IKS)	Generic IKS	3	2	50	0	50
Co-Curricular Courses (CC)	Co-Curricular Courses (CC)	Physical Education – I	@ Department	2	50	0	50
		Total	-	22	270	280	550

FYBBBA-CA Semester II							
Course Type	Course	Paper Title	Hours / Week	Credits	Internal	External	Total
Major Mandatory (06)	Major Mandatory 4	Advance C Programming	3	2	15	35	50
	Major Mandatory 5	Relational Database Management System (RDBMS)	3	2	15	35	50

	Major Mandatory 6 (Practical)	Computer Laboratory based on Advance C and RDBMS	5	2	15	35	50
Minor	Minor 1	Principle and Practices of Management	3	2	15	35	50
Open Elective (OE)	Open Elective 3	Introduction to Data Science	3	2	15	35	50
	Open Elective 4	Tally Prime	3	2	50	0	50
Vocational Skill Development Course (VSC)	Vocational Skill Development Course (VSC) (Practical)	Web Technology	5	2	15	35	50
Skill Enhancement Course (SEC)	Skill Enhancement Course (SEC)	E-Commerce	3	2	15	35	50
Ability Enhancement Course (AEC)	Ability Enhancement Course (AEC)	Business Communication Skills-II	3	2	15	35	50
Value Education Course (VEC)	Value Education Course (VEC)	Democracy Awareness & Gender Sensitization	3	2	50	0	50
Co-Curricular Courses (CC)	Co-Curricular Courses (CC)	Physical Education – II	@ Depar tment	2	50	0	50
		Total	-	22	270	280	550

SYBBBA-CA Semester III

Course Type	Course	Paper Title	Hours / Week	Credits
Major Mandatory (08)	Major Mandatory 7	Data Structure	**5**	4
	Major Mandatory 8	PHP	**5**	4
Minor	Minor 2 (Practical)	Computer Laboratory based on DS, PHP	**5**	4
Open Elective (OE)	Open Elective 5	To be selected from the basket of the other faculty	**3**	2
Vocational Skill Development Course (VSC)	Vocational Skill Development Course (VSC) (Practical)	Web development tools	**5**	2
Ability Enhancement Course (AEC)	Ability Enhancement Course (AEC)	Modern Indian Languages 1	**3**	2
Field Projects (FP)	Project	Project based on Web Applications	**5**	2
Co-Curricular Courses (CC)	Co-Curricular Courses (CC)	NSS/NCC/Yoga Education/Health and Wellness/Fine Arts-I	@ Depar tment	2
		Total	-	22

SYBBBA-CA Semester IV

Course Type	Course	Paper Title	Hours / Week	Credits
Major Mandatory (08)	Major Mandatory 9	Object Oriented Programming using C++	5	4
	Major Mandatory 10	Advance PHP	5	4

Minor	Minor 3 (Practical)	Computer Laboratory based on CPP, Adv PHP	5	4
Open Elective (OE)	Open Elective 6	Digital Marketing	5	2
Skill Enhancement Course (SEC)	Skill Enhancement Course (SEC)	Computer Network	3	2
Ability Enhancement Course (AEC)	Ability Enhancement Course (AEC)	Modern Indian Languages 2	3	2
Field Projects	Project	Project based on Digital Marketing	5	2
Co-Curricular Courses (CC)	Co-Curricular Courses (CC)	NSS/NCC/Yoga Education/Health and Wellness/Fine Arts-II	@ Depar tment	2
		Total	-	22

TYBBBA-CA Semester V

Course Type	Course	Paper Title	Hours / Week	Cred its
Major Mandatory (10)	Major Mandatory 11	Java Programming	5	4
	Major Mandatory 12	Mobile Application Development	5	4
	Major Mandatory 13 (Practical)	Computer Laboratory based on Java and Mobile Application Development	3	2
Major Elective	Major Elective 1	Linux Operating System	5	4
Minor	Minor 4	Software Engineering	5	4
Vocational Skill Development Course (VSC)	Vocational Skill Development Course (VSC) (Practical)	Dot Net Programming	5	2
Field Projects (FP)/ Community Engagement and Service corresponding to the Major (CEP)	Project	Project based on Mobile Application Development	5	2
		Total	-	22

TYBBBA-CA Semester VI

Course Type	Course	Paper Title	Hours / Week	Cred its
Major Mandatory (10)	Major Mandatory 14	Recent Trends in IT	5	4
	Major Mandatory 15	Python	5	4
	Major Mandatory 16 (Practical)	Computer Laboratory based on Python	5	2
Major Elective	Major Elective 2	Internet of Things	3	2
	Major Elective 3	Software Testing	3	2
Minor	Minor 5	Management Information Systems	5	4

On Job Training (OJT)	On Jot Training	Internship + Project	After the final exams of Sem V	4
		Total	-	22

Detail Syllabus

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures per week
I	BBACA101T	Major Mandatory	Problem Solving Using C	02	03

Course Objectives:

1. To introduce the foundations of computing, programming and problem- solving using computers.
2. To develop the ability to analyze a problem and devise an algorithm to solve it.
3. To formulate algorithms, pseudocodes and flowcharts for arithmetic and logical problems
4. To understand structured programming approach.
5. To develop the basic concepts and terminology of programming in general.
6. To implement algorithms in the 'C' language.
7. To test, debug and execute programs.

Course Outcome:

At the end of the course, students will be able to

CO1	1. Define algorithms and explain their characteristics
CO2	2. Formulate algorithm and draw flow chart to solve a given problem
CO3	3. Explain use of appropriate data types, control statements
CO4	4. Demonstrate ability to use top-down program design

Syllabus

Unit	Title and Contents	No. of Lectures
1	**'C' Fundamentals** History of 'C' language. Application areas. Structure of a 'C' program. 'C' Program development life cycle. Function as building blocks. 'C' tokens Character set, Keywords, Identifiers Variables, Constants (character, integer, float, string, escape sequences, enumeration constant). Data Types (Built-in and user defined data types). Operators, Expressions, types of operators, Operator precedence and Order of evaluation. Character input and output. String input and output. Formatted input and output **Control Structures**	**15**

	Decision making structures: - if, if-else, switch and conditional operator. Loop control structures: - while do while, for. Use of break and continue. Nested structures. Unconditional branching (goto statement)	
2	**Functions** Concept of function, Advantages of Modular design. Standard library functions. User defined functions: - declaration, definition, function call, parameter passing (by value), return statement. Recursive functions. Scope of variables and Storage classes. **Arrays** Concept of array. Types of Arrays – One, Two and Multidimensional array. Array Operations - declaration, initialization, accessing array elements.	**15**

Reference Books

1. How to Solve it by Computer, R.G. Dromey, Pearson Education.
2. Problem Solving and Programming Concept, Maureen Sprankle,7th Edition, Pearson Publication.
3. 3C: the Complete Reference, Schildt Herbert, 4 th edition, McGraw Hill
4. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India
5. The ‘C’ programming language, Brian Kernighan, Dennis Ritchie, PHI
6. Programming in C, A Practical Approach, Ajay Mittal, Pearson
7. Programming with C, B. Gottfried, 3rd edition, Schaum’s outline Series, Tata McGraw Hill.
8. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill.

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures per week
I	BBACA102T	Major Mandatory	Database Management System	02	03

Course Objectives:

1. To make students understand the concept of Database Management System
2.To develop Database application

Course Outcome:

CO1	To understand the basic concepts and the applications of database systems.
CO2	To formulate Queries using SQL and Relational Formal Query Languages

Unit	Title and Contents	No. of Lectures
1	**Introduction to Databases Management and Data Models** 1.1 Introduction 1.2 Application Of DBMS 1.3 Advantages of DBMS 1.4 Users of DBMS 1.4.1 Database Designers 1.4.2 Application Programmer 1.4.3 Sophisticated Users 1.4.5 End Users 1.5 Views of Data 1.6 Data Models 1.6.1 Relational Model 1.6.2 Network Model 1.6.3. Hierarchical Model 1.7 Entity Relationship Diagram (ERD) 1.8 Features of ERD 1.9 Cases Studies on ER Model 1.10 Introduction to Relational Model 1.11 Basic Concepts: Relation, tuple, attribute 1.12 Key: Super Key, Candidate Key, Primary Key, Foreign Key	15
2	**SQL (Structured Query Language)**	15

	2.1 Introduction 2.2 Normalization 2.2.1 First Normal Form0 2.2.2 Second Normal Form 2.2.3 Third Normal Form 2.2.4 Boyce - Codd Normal Form 2.2 Basic Structure 2.3 DDL Commands 2.4 DML Commands 2.5 Simple Queries 2.6 Constraint (Not NULL, Check, Unique, Default) 2.7 Aggregate function (Min, Max, Avg, Count, Sum) 2.8 Clause (Group By, Order By, Having) 2.9 Nested Queries 2.10 Case Study on SQL	

Reference Books:

1. Database System Concepts by Henry Korth and A. Silberschatz
2. SQL, PL/SQL The Programming Language Oracle: - Ivan Bayross, BPB Publication.
3. Database Systems Concepts, Designs and Application by Shio Kumar Singh, Pearson
4. Introduction to SQL by Reck F. van der Lans by Pearson
5. Modern Database Management by Jeffery A Hoffer, V. Ramesh, Heikki Topi, Pearson
6. Database Management Systems by Debabrata Sahoo, Tata McGraw Hill

Semester No.	Course Code	Type of Course	Course Title	Credits	Lecture Hours/Week
I	OE-103-MTS	Open Elective	Business Mathematics – I	2	3

Note: This course is taken from OE basket of Faculty of Science and Technology.

Course Objectives	
1.	To provide solid Mathematical Foundation for BBA Students in Business and Finance.
2.	To help the students for various mathematical topics with Practical Business Application.
3.	To enhance problem – solving Skills and ability for Academic and Professional Success.
4.	To make students understands mathematics behind commerce and Management.
5.	To foster conceptual Clarity and Confidence in Mathematical Competence.

Course Outcome The student will be able to	
CO1	understand the Concepts of Ratio, Proportion, Percentage and Partnership.
CO2	apply the mathematical concepts to solve real-world financial problems.
CO3	understand the equated monthly instalments (EMI) for loans and mortgages.
CO4	apply the simple and compound interest for various financial instruments.
CO5	analyze models related to Finance and can solve them.
CO6	remember the computation of Dividend and Return on Investment in shares.

Unit	Title and Contents	No. of Lecture Hours
1	Ratio, Proportion, Percentage 1.1 Introduction to Ratios and Proportions, Applications of Ratios and Proportions, Percent- ages and its applications. 1.2 Concept of Commission and Brokerage, Types of Commission, Partnership, Practical applications.	**15**
2	Interest and Dividend 2.1 Simple interest and compound Interest. 2.2 Equated Monthly Instalments (EMI), EMI on reducing balance, EMI on at and floating rate of interest. 2.3 Concept of shares and dividends, Types of Shares, Problems on dividend and return on investment on shares.	**15**

References

1. Practical Business Mathematics by S. A. Bari, New Literature Publishing Company, New Delhi, India.
2. Mathematics for Commerce by K. Selvakumar, Notion Press, Chennai, India.
3. Business Mathematics with Applications by Dinesh Khattar and S. R. Arora, S. Chand Publishing, New Delhi, India.
4. Fundamentals of Business Mathematics by M. K. Bhowal, Asian Books Pvt. Ltd, New Delhi.
5. Business Mathematics by D.C. Sancheti and V. K. Kapoor, Sultan Chand and Sons. 6. Business Mathematics by J. K. Singh, Himalaya Publishing House.

Semester No.	Course Code	Type of Course	Course Title	Credits	Lecture Hours/Week
I	OE-103-STS	Open Elective	Business Statistics – I	2	3

Note: This course is taken from OE basket of Faculty of Science and Technology

Course Objectives	
1.	To understand role and importance of statistics in various business situations
2.	To develop skills related with basic statistical technique
3.	To learn some elementary statistical methods for data collection, presentation and analysis of data.
4.	To develop right understanding regarding data interpretation
5.	To familiarize the students with applications of Statistics in Business and Management

Course Outcome	
CO1	understand basic concepts in statistics
CO2	collect, present, analyze and interpret the data and graphs
CO3	deal data in business problems
CO4	evaluate feasibility business problems using statistical techniques
CO5	prepare business report using various statistical techniques

Unit	Title and Contents	No. of Lecture Hours
1	Frequency Distribution 1.1 Raw data, variable, discrete variable, continuous variable, constant, attribute with illustration. 1.2 Classification- Concept and definition of classification, objectives of classification, types of classification. 1.3 Frequency Distribution- Discrete and Continuous frequency distribution, Cumulative frequency and Cumulative frequency distribution.	10

	1.4 Graphs & Diagram- Histogram, Ogive curve, Pie-Diagram, Bar Diagram, Multiple bar Diagram, Sub-divided bar diagram	
2	Measure of Central Tendency 2.1 Concept and meaning of Measure of Central Tendency, Objectives of Measure of Central Tendency, Requirements of good Measure of Central Tendency. 2.2 Types of Measure of Central Tendency, Arithmetic Mean (A.M), Median, Mode for discrete and Continuous frequency distribution, Merits & Demerits of A.M., Median, Mode, Numerical Problem. 2.3 Determination of Mode and Median graphically. 2.4 Empirical relation between mean, median and mode. 2.5. Combined Mean 2.6. Numerical Problems	**10**
3	Measure Dispersion Concept of Dispersion, Measures of Dispersion - Range, Variance and Standard Deviation (S.D.) for Grouped and ungrouped data, Measures of relative dispersion- Coefficient of range and coefficient of Variation, Examples.	**10**

Sr. No	Title of the Book	Author/s	Publication	Place
1.	Business Mathematics and Statistics -I	Dr. M. P. Waghmare	Thakur Publication	Pune
2.	Business Statistics	Girish Phatak	Tech – Max	Pune
3.	Statistics for Business	Dr. S. K. Khandelwal	International Book House	New Delhi
4.	Fundamentals of Business Statistics	J.K. Sharma	Pearson	New Delhi
5.	Business Statistics	G.C. Beri	The McGraw-Hill companies	New Delhi

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures Hours per week
I	BBACA101VSC	Vocational Skill Development Course (VSC)	Office Automation tools	02	03

Course Objective:

To make students understand and learn various Office Automation Tools like MSWord, MS Excel & MS PowerPoint.

Course outcome:

CO1	The students will be able to apply various Office Automation Tools - MSWord, MS Excel & MS PowerPoint
CO2	Use of modern office equipment in business

Unit	Title and Contents	No. of Lectures
1.	**Introduction** Concept of Windows, Icon, Menu Desktop Creating Folders and Shortcuts Finding Files& Folders Creating, Copying, Moving and Deleting files Windows Explorer Basic DOS Commands **Word Processing Package** Typing, Editing, Proofing &reviewing Formatting text &Paragraph Automatics Formatting and Styles Working with Tables Graphics and Frames Mail Merge	15
2.	**Spread sheet package** Concept of worksheet Working& Editing in Workbooks Creating Formats & Links Protecting and Hiding data Built in Functions (Mathematical, Statistical, String &Date) Formatting a Worksheet & Creating graphics objects Creating Charts (Graphics), Formatting and analyzing data Organizing Data in a List (Data Management) Sharing & Importing Data Printing **Presentation Package** Creating and Editing Slides Creating and Editing objects in the slide Animation Creating and Running Slideshow Templates	15

Reference Books:

1. EXCEL2007 Made Simple by Satish Jain, BPB
2. Word2007 by Rutkosky, BPB
3. PowerPoint2007 Made Simple by Satish Jain, BPB
4. MasteringEXCEL4forWindows-Chester–BPB
5. MicrosoftOfficeWord2007 Plain & Simple, Joyce & Moon, PHI
6. MicrosoftOfficeExcel2007Plain&Simple, Frye, PHI
7. MicrosoftOfficePowerPoint2007Plain&Simple, Muir, PHI
8. 2007MicrosoftOfficeSystemPlain&Simple, Joyce Moon, PHI
9. EXCEL5forWindowsQuick&Easy-JonesTECH
10. Excel Functions & formulas by Bernd Held, BPB
11. MasteringWindows2000Cowat-BPB
12. MSOFFICE2007-TRAININGGUIDEbySatishJain, BPB
13. Internet: An Introduction Cisiems–Tata Mac, D. Boody–BPB
14. Internet 6 in 1–Joe Krayuak & Harbraken, PHI
15. Internet access essential–Tittle & M. Robbins, AP professional PCSoftwareforWindows2003 Made Simple, RKTaxali, TMH

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures Hours per week
I	**BBACA101SEC**	**Skill Enhancement Course (SEC)**	**Programming Principles and Algorithm**	**02**	**03**

Course Objectives:

1. To make students understand the concept of Algorithm and Flowchart.
2. To develop Analytical / Logical Thinking and Problem-Solving capabilities
3. To Know the Basics of Programming.

Course Outcome:

CO1	To understand how to use programming in day-to-day Applications
CO2	To apply skills of algorithm and flowchart to solve the businesses problem

Unit	Title and Contents	No. of Lectures
1	**Introduction** 1 Concept: Problem solving, Program development cycle 2 Algorithm, Characteristics of an algorithm 3 Flowcharts 4 Simple Examples: Algorithms and flowcharts 4.1 Addition / Multiplication of integers 4.2 Determining if a number is +ve / -ve / even / odd 4.3 Maximum of 2 numbers, 3 numbers 4.4 Sum of first n numbers, given n numbers, Digit reversing, Palindrome number, Armstrong number 4.5 Table generation for n, Factorial, Prime number, Factors of a number etc. (Write algorithms and draw flowcharts)	**15**
2	**Recursion** 1.1 Concept: Multiplication, Factorial, Fibonacci series, Permutation Generation 1.2 Algorithms using arrays Maximum and minimum of array, reversing elements of an array. 1.3 Mean and Median of n numbers 1.4 Row major and Column major form of array representation 1.5 Matrices: Addition, Multiplication, Transpose, upper/lower triangular	**15**

Reference Books:

1. Let us C-Yashwant Kanetkar.
2. Programming in C- Balagurusamy
3. How to solve it by Computer – R. G. Dromy
4. Introduction to algorithms – Cormen, Leiserson, Rivest, Stein

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures per week
I	BBACA101AEC	Ability Enhancement Course (AEC)	Business Communication Skills-I	02	03

Course Objectives:

1. To understand what the Need and Significance of communication in personal and business world
2. To understand system of communication and their utility
3.

Course Outcome:

Student will able

CO1	To understand the concept, process, and importance of communication
CO2	To apply gain knowledge of media of communication in businesses
CO3	To develop skills of effective communication - both written and oral

Unit	Title and Contents	No. of Lectures
1	**Introduction** 1.1Meaning, Definition of Communication 1.2 Need for effective communication 1.3 Process of Communication 1.3 C's of effective communication, 1.4 Types of Communication- 1.4.1 Verbal communication- Formal and Grapevine, 1.4.2 Nonverbal communication: -Gestures, Postures, Facial Expression, Eye Contacts, Body Language (Kinesics), Silence, Tips for Improving Non-Verbal Communication 1.5 Barriers to communication 1.6 over comings barriers to communication 1.7 Listening Skills- Types of Listeners, Tips to be good listener. 1.8 Different Media of Communication- E-mails, social media, Fax communication, Video Conferencing, Blogs	15
2	**Writing Skills** 2.1Written Communication-Merits and Merits 2.2. Report Writing- Meaning Definition of Report Importance of good report, Qualities of a good report, Tips for writing good report 2.3 Email Correspondence - Writing effective emails 2.4 Appropriate email subject lines 2.5 Email etiquette and conventions 2.6 Practice writing and receiving emails 2.7 Business Letters – Structure and Components of Business letters, Drafting Business letters	15

References

1. Business Communication, R.K. Madhukar, Vikas Publishing House
2. Business Communication, Homai Pradhan, N.S. Pradhan, Himalaya Publishing House
3. Business Communication, K.K. Sinha, Taxman Publications

Semester No.	Course Code	Type of Course	Course Title	Credits	Lectures per week
I	**BBACA101VEC**	**Value Education Course (VEC)**	**Environmental Awareness**	**02**	**03**

Course Objectives:

1) To provide an opportunity to acquire the knowledge, values, attitudes, commitment, and skills needed to protect and improve the environment
2) To develop conscious towards a cleaner and better managed environment

Course Outcome:

Student will able

CO1	To understand Environmental pollution.
CO2	To apply and promote green practices at home and at work

Unit	Title and Contents	No. of Lectures
1	Introduction - Environmental studies Definition, scope importance and need for public awareness. (Multidisciplinary nature of environmental studies) 2 Environmental Pollution -Definition, Causes, effects on human, water, soil, air (Mother Earth) Air pollution, Water pollution, Soil pollution Marine pollution, Noise pollution, Thermal pollution, nuclear hazards	15
2	Various Government initiatives for conservation of Environment. Controlling measures), Solid waste Management: Causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Pollution case studies. Disaster management: floods, earthquakes, cyclone and landslides.	15

S.Y.B.B.A.(C.A.) Semester –III

Course Code: CA-301

Subject: Digital Marketing

Objectives:

1. The aim of this syllabus is to give knowledge about using digital marketing in and as business.

2. To make SWOT analysis, SEO optimization and use of various digital marketing tools.

Unit	Topic	No. of Lectures
1.	**E-Commerce** 1.1 Introduction 1.2 Understanding Internet Marketing 1.3 Search Engine Optimization 1.4 Search Engine Marketing 1.5 Email Marketing 1.6 Digital Display Marketing	**4**
2.	**Introduction to New Age Media (Digital) Marketing** 2.1 What is Digital Marketing 2.2 Digital vs. Real Marketing 2.3 Digital Marketing Channels 2.4 Types of Digital Marketing(Overview)-Internet Marketing ,Social Media Marketing, Mobile Marketing	**4**
3.	**Creating Initial Digital Marketing Plan** 3.1 Content management 3.2 SWOT analysis: Strengths, Weaknesses, Opportunities, andThreats 3.3 Target group analysis EXERCISE: Define a target group	**4**
4.	**Marketing using Web Sites** 4.1 Web design 4.2 Optimization of Web sites 4.3 MS Expression Web EXERCISE: Creating web sites, MS Expression	**4**
5.	**Search Engine Optimization** 5.1 SEO Optimization 5.2 Writing the SEO content EXERCISE: Writing the SEO content	**4**
6.	**Customer Relationship Management** 6.1 Introduction to CRM 6.2 CRM platform 6.3 CRM models EXERCISE: CRM strategy	**4**

7.	**Social Media Marketing** 7.1 Understanding Social Media Marketing 7.2 Social Networking (Facebook, Linkedin, Twitter, etc.) Social Media (Blogging, Video Sharing - Youtube, Photosharing – Instagram, Podcasts) 7.3 Web analytics - levels 7.4 Modes of Social Media Marketing- 7.4.1 **Creating a Facebook page** Visual identity of a Facebook page , Types of publications, Facebook Ads , Creating Facebook Ads , Ads Visibility 7.4.2 **Business opportunities and Instagram options** Optimization of Instagram profiles , Integrating Instagram with a Web Site and other social networks ,Keeping up with posts 7.4.3 **Business tools on LinkedIn** Creating campaigns on LinkedIn , Analyzing visitation on LinkedIn 7.4.4 **Creating business accounts on YouTube**YouTube ,Advertising , YouTube Analytics 7.4.5 **E-mail marketing** E-mail marketing plan , E-mail marketing campaign analysis , Keeping up with conversions 7.5 Digital Marketing tools: Google Ads, FaceBook Ads, Google Analytic, Zapier, Google Keyword Planner EXERCISE: Social Media Marketing plan. EXERCISE: Making a Facebook page and Google Ads	**1** **2** **2** **3** **3** **3** **3** **3** **(20)**
8.	**Digital Marketing Budgeting** 8.1 Resource planning 8.2 Cost estimating 8.3 Cost budgeting 8.4 Cost control	**4**
Total		**48**

Reference Books:

1) Digital Marketing for Dummies By Ryan Deiss and Russ Hennesberry
2) Advertising and Promotion: An Integrated Marketing Communications Perspective, George Belch, San Diego University Michael Belch, San Diego University
3) Advertising Management: Rajeev Batra, John G. Myers, David A. Aaker
4) Belch: Advertising & Promotions (TMH)
5) The Social Media Bible: Tactics, Tools, & Strategies for Business Success by Lon Safko
6) Web Analytics 2.0 – AvinashKaushik

S.Y.B.B.A(C.A) Semester – III

Course Code: CA-302

Subject : Data Structure

Objectives:

1. To understand the concepts of ADTs
2. To learn linear data structures – lists, stacks, and queues
3. To understand sorting, searching and hashing algorithms
4. To apply Tree and Graph structures

Unit	Topic	No. of Lectures
1	**Basic Concept and Introduction to Data Structure** 1.1 Pointers and dynamic memory allocation 1.2 Algorithm-Definition and characteristics 1.3 Algorithm Analysis -Space Complexity -Time Complexity - Asymptotic Notation Introduction to Data structure 1.4 Types of Data structure 1.5 Abstract Data Types (ADT) Introduction to Arrays and Structure 1.6 Types of array and Representation of array 1.7 Polynomial - Polynomial Representation - Evaluation of Polynomial - Addition of Polynomial 1.8 Self Referential Structure	5
2	**Linear data structures** 2.1 Introduction to Arrays - array representation 2.2 Sorting algorithms with efficiency - Bubble sort, Insertion sort, Merge sort, Quick Sort, Selection Sort 2.3 Searching techniques –Linear Search, Binary search	6
3	**Linked List** 3.1 Introduction to Linked List 3.2 Implementation of Linked List – Static & Dynamic representation, 3.3 Types of Linked List - Singly Linked list(All type of operation) - Doubly Linked list (Create , Display) - Circularly Singly Linked list (Create, Display) - Circularly Doubly Linked list (Create, Display) 3.4 Generalized linked list – Concept and Representation	6
4	**Stacks** 4.1 Introduction 4.2 Representation- Static & Dynamic 4.3 Primitive Operations on stack 4.4 Application of Stack 4.5 Conversion of Infix, prefix, postfix , Evaluation of postfix and prefix	8

	4.6 Simulating recursion using stack	
5	**Queues** 5.1 Introduction 5.2 Representation - Static & Dynamic 5.3 Primitive Operations on Queue 5.4 Circular queue, priority queue 5.5 Concept of doubly ended queue	**4**
6	**Trees** 6.1 Concept & Terminologies 6.2 Binary tree, binary search tree 6.3 Representation – Static and Dynamic 6.4 Operations on BT and BST – create, Insert, delete, , counting leaf, non-leaf & total nodes , 6.5 Tree Traversals (preorder, inorder, postorder) 6.6 Application - Heap sort 6.7 Height balanced tree- AVL trees- Rotations, AVL tree examples.	**12**
7	**Graph** 7.1 Concept & terminologies 7.2 Graph Representation – Adjacency matrix, adjacency list, inverse Adjacency list, adjacency multilist, orthogonal list 7.3 Degree of Graph 7.4 Traversals – BFS and DFS 7.5 Applications – AOV network – topological sort, AOE network – criticalPath	**7**
Total		**48**

Reference Books:

1. Fundamentals of Data Structures ---- By Horowitz Sahani (Galgotia)
2. Data Structures using C and C++ --- By YedidyahLangsam, Aaron M. Tenenbaum, Moshe J. Augenstein
3. Introduction to Data Structures using C---By Ashok Kamthane
4. Data Structures using C --- Bandopadhyay&Dey (Pearson)
5. Data Structures using C ---By Srivastava BPB Publication.

S.Y.B.B.A. (C.A.) Semester –III

Course Code: CA-303

Subject: Software Engineering

Objectives:

1. To understand System concepts.
2. To understand Software Engineering concepts.
3. To understand the applications of Software Engineering concepts and Design in Software development

Unit	Topic	No. of lectures
1	**Introduction to System Concepts** 1.1 Definition 1.2 Basic Components 1.3 Elements of the System 1.4 Types of System 1.5 System Characteristics	4
2	**Introduction to Software Engineering** 2.1 Definition of Software 2.2 Characteristics of Software 2.3 Definition of Software Engineering 2.4 Need for Software Engineering 2.5 Mc Call's Quality factors 2.6 The Software Process 2.7 Software Product and Process 2.8 V& V Model	6
3	**Software Development Life Cycle** 3.1 Introduction 3.2 Activities of SDLC 3.3 A Generic Process Model 3.4 SDLC 3.5 Waterfall Model 3.6 Incremental Process Models 3.7 Prototyping Model 3.8 Spiral Model	8
4	**Requirement Engineering** 4.1 Introduction 4.2 Requirement Elicitation 4.3Requirement Elaboration 4.4 Requirement Gathering 4.5 Feasibility study	8

	4.6 Fact Finding Techniques 4.7 SRS Format	
5	**Analysis And Design Tools** 5.1 Decision Tree and Decision Table 5.2 Data Flow Diagrams (DFD) (Up to 2nd level) 5.3 Data Dictionary 5.4 Elements of DD 5.5 Advantages and Disadvantages of DD 5.6 Input and Output Design 5.7 Structured Design Concepts 5.8 Structure Chart 5.9 Coupling and Cohesion 5.10 Compulsory Case Studies on above topics	**12**
6	**Software Testing** 6.1 Definition 6.2 Software testing Process 6.3 Unit Testing 6.4 Integration Testing 6.5 System Testing	**6**
7	**Software Maintenance and Software Re-Engineering** 7.1 Maintenance definition and types 7.2 Software reengineering 7.3 Reverse Engineering 7.4 Restructuring and forward Engineering.	**4**
Total		**48**

Reference Books:

1. Software Engineering: A Practitioner's Approach- Roger S. Pressman, McGraw hill International Editions 2010(Seventh Edition)

2. System Analysis, Design and Introduction to Software Engineering (SADSE) - S. Parthsarthy, B.W. Khalkar

3. Analysis and Design of Information Systems(Second Edition) - James A. Senn, McGraw Hill

4. System Analysis and Design- Elias Awad, Galgotia Publication, Second Edition

S.Y.B.B.A.(C.A.) Semester – III

Course Code: CA- 304 (Option)

Subject: Angular - JS

Objectives:

- By the end of this course, the students should be able to Understand Client Side MVC and SPA
- Explore AngularJS Component
- Develop an AngularJS Single Page Application
- Create and bind controllers with Javascript
- Apply filter in AngularJS application

Unit	Topics	No. of Lectures
1	**AngularJS Core Concepts:** 1.1 What is AngularJS? 1.2 Difference between Javasript and Angular JS 1.3 Advantages of Angular 1.4 AngularJS MVC Architecture 1.5 Introduction to SPA 1.6 Setting up the environment 1.7 First App using MVC architecture	**8**
2	**AngularJS Directives and Expressions:** 2.1 Understanding ng attributes ng-app, ng-init, ng-model, ng-controller, ng-bind, ng-repeat, ng-show, ng-readonly, ng-disabled, ng-if, ng-click 2.2 Expression and Data Binding 2.3 Working with directives	**10**
3	**AngularJS Modules, Controller, View and Scope:** 3.1 Angular Modules 3.2 Angular Controller 3.3 Angular View 3.4 Scope hierarchy	**10**
4	**Filter, Forms and Ajax Filters** 4.1 Built-in filters - upper case and lower case filters, date ,currency and number formatting ,orderBy, filter ,custom filter, 4.2 Angular JS Forms – Working with AngularJS forms, model binding,	**12**

	form controller ,Using CSS classes, form events , custom model update triggers ,custom validation, $http service , 4.3 Ajax implementation using $http	
5	**Dependency Injection, Services** 5.1 What is dependency injection? 5.2 Understanding services 5.3 Using built-in service 5.4 Creating custom service, 5.5 Injecting dependency in service	**8**
Total		**48**

Reference Books:

1. Beginning Angular with Typescript (updated to Angular 5) by Greg Lim
2. Mastering Web Application Development with AngularJS by Pawel Kozlowski, Peter Bacon Darwin
3. https://www.tutorialsteacher.com/angularjs/angularjs-scope

S.Y.B.B.A.(C.A.) Semester – IV

Course Code: CA- 304(Option)

Subject: PHP

Objectives:

1. Understand how server-side programming works on the web.
2. Using PHP built-in functions and creating custom functions
3. Understanding POST and GET in form submission.
4. How to receive and process form submission data.
5. Read and process data in a MySQL database.

Unit	Topic	No. of Lectures
1	**PHP Basics** 1.1 Setting up a development environment 1.2 Variables, numbers and strings 1.3 Calculations with PHP 1.4 Using Arrays	6
2	**Control Structures and Loops** 2.1 Conditional Statements 2.2 Using Loops for Repetitive tasks 2.3 Combing Loops and Arrays	7
3	**Functions, Objects and Errors** 3.1 PHP's Built-in functions 3.2 Creating Custom functions 3.3 Passing Values by Reference 3.4 Understanding Objects	7
4	**Working with Forms** 4.1 Building a Form 4.2 Processing a Form's Data 4.3 Differences between POST and GET 4.4 Preserving User Input	7
5	**More with Forms** 5.1 Dealing with checkboxes and radiobuttons 5.2 Retrieving values from lists 5.3 Validating and restricting data 5.4 Sending Email	7
6	**Storing and Protecting Data** 6.1 Setting and Reading Cookies 6.2 Protecting Online Files 6.3 Understanding Session Variables	7
7	**MySQL Database Overview**	7

	7.1 phpMyAdmin Overview 7.2 Using a MySQL Database 7.3 Reading and Writing Data	
Total		**48**

Reference Books:

1. Php: A Beginner's Guide 1st EditionMcGraw-Hill Osborne Media; 1 edition by VikramVaswani
2. Murach's PHP and MySQL (2nd Edition)by Joel Murach and Ray Harris
3. PHP: The Complete Reference Paperback – 1 Jul 2017by Steven Holzner (Author)

S.Y.B.B.A.(C.A.) Semester – III

Course Code: CA- 305(Option)

Course Title : Big Data

Objectives:

1. To enable learners to develop expert knowledge and analytical skills in current and developing areas of analysis statistics, and machine learning

2. To enable the learner to identify, develop and apply detailed analytical, creative, problem solving skills.

3. Provide the learner with a comprehensive platform for career development, innovation and further study.

Unit	Topic	No. of lectures
1	**INTRODUCTION TO BIG DATA** 1.1 Introduction to Big Data 1.2 Types of Digital Data 1.3 Big Data Analytics 1.4 Application of Big data	5
2	**INTRODUCTION TO DATA SCIENCE** 2.1 Basics of Data Analytics 2.2 Types of Analytics – 2.2.1 Descriptive, 2.2.2 Predictive, 2.2.3 Prescriptive 2.2.4 Statistical Inference 2.3 Populations and samples 2.3.1 Statistical modelling, 2.3.2 Probability 2.3.3 Distribution 2.3.4 Correlation 2.3.5 Regression	10
3	**INTRODUCTION TO MACHINE LEARNING** 3.1 Basics of Machine Leaning 3.2 Supervised Machine Learning 3.2.1 K- Nearest-Neighbours, 3.2.2 Naïve Bayes 3.2.3 Decision tree 3.2.4 Support Vector Machines	20

	3.3 Unsupervised Machine Learning 3.3.1 Cluster analysis 3.3.2 K means 3.3.3 EM Algorithm 3.3.4 Association Rule Mining 3.3.5 Apriori algorithms 3.4 Regression Analysis 3.4.1 Linear Regression 3.4.2 Nonlinear Regression	
4	**DATA ANALYTICS WITH R/ WEKA MACHINE LEARNING** 4.1 Introduction 4.2 Data Manipulation 4.3 Data Visualization 4.4 Data Analysis	**13**
Total		**48**

Reference Books:

1. SeemaAcharya, SubhasiniChellappan, "Big Data Analytics" Wiley 2015.
2. Jay Liebowitz, “Big Data and Business Analytics” Auerbach Publications, CRC press (2013)
3. ArvindSathi, “BigDataAnalytics: Disruptive Technologies for Changing the Game”, MC Press, 2012

S.Y.B.B.A.(C.A.) Semester – III

Course Code: CA-305 (Option)

Course Title : BlockChain

PREREQUISITES:

This course is highly technical in nature and would require the student to be comfortable with coding. To prepare for the class all students MUST:

- Understanding of basic programming language like Java, or Javascript.
- Understanding of PKI and Docker.

WHAT YOU'LL LEARN

- Understand what and why of Blockchain
- Explore the major components of Blockchain
- Learn about Bitcoin, Cryptocurrency, Ethereum
- Deploy and exercise example smart contracts
- Identify a use case for a Blockchain application
- Create your own Blockchain network application

COURSE OBJECTIVES

By the end of the course, students will be able to

1. Understand how blockchain systems (mainly Bitcoin and Ethereum) work,
2. To securely interact with them,
3. Design, build, and deploy smart contracts and distributed applications,
4. Integrate ideas from blockchain technology into their own projects.

Unit	Topic	No. of Lectures
1	**Introduction To Blockchain** 1.1 Digital Trust 1.2 Asset 1.3 Transactions 1.4 Distributed Ledger Technology 1.5 Types of network 1.6 Components of blockchain or DLT 1.7 Ledger 1.7.1. Blocks 1.7.2. Blockchain 1.8 PKI and Cryptography 1.8.1. Private keys 1.8.2. Public keys 1.8.3. Hashing 1.8.4. Digital Signature 1.9. Consensus	12

	1.9.1. Byzantine Fault 1.9.2. Proof of Work 1.9.3. Poof of Stake 1.10. Security 1.10.1.DDos 1.11 Cryptocurrency 1.12.Digital Token	
2.	**How Blockchain Works** 2.1 How Blockchain Works 2.2. Structure of Blockchain 2.3.Block 2.4. Hash 2.5. Blockchain 2.6. Distributed 2.7. Lifecycle of Blockchain 2.8. Smart Contract 2.9. Consensus Algorithm 2.10 Proof of Work 2.11 Proof of Stake 2.12 Practical Byzantine 2.13 Fault Tolerance 2.14 Actors of Blockchain 2.15 Blockchain developer 2.16 Blockchain operator 2.17 Blockchain regulator 2.18 Blockchain user 2.19 Membership service provider 2.20 Building A Small Blockchain Application	**12**
3.	**Introduction to Bitcoin** 3.1 Currency 3.2 Double Spending 3.3 Cryptocurrency 3.4 P2P Payment Gateway 3.5 Wallet 3.6 Mining	**8**
4.	**Ethereum** 4.1.Ethereum network 4.2. EVM 4.3.Transaction fee 4.4.Mist 4.5.Ether, gas 4.6.Solidity - Smart contracts 4.7.Truffle 4.8.Web3 4.9.Design and issue Cryptocurrency 4.10. Mining	**8**

	4.11. DApps 4.12. DAO	
5	**Introduction To Hyperledger Fabric V1.1** 5.1. Introduction to Hyperledger 5.2 What is Hyperledger 5.3 Why Hyperledger 5.4 Where can Hyperledger be used 5.5 Hyperledger Architecture 5.6 Membership 5.7 Blockchain 5.8 Transaction 5.9 Chaincode 5.10 Hyperledger Fabric 5.11 Features of Hyperledger	**8**
Total		**48**

References:

Text Book

1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder,

Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press (July 19, 2016).

Reference Books

1. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies
2. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System
3. DR. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger,"Yellow paper.2014.
4. Nicola Atzei, Massimo Bartoletti, and TizianaCimoli, A survey of attacks on Ethereum smart contracts

SavitribaiPhule Pune University
Syllabus for BBA(CA) (CBCS 2019 Pattern)
Details for Skill Enhancement (Add-On) Courses

AECC - Course Title: - (M)Basic Course in Environmental Awareness
Credit -2 & Hours -30

Objectives:

1) To provide an opportunities to acquire the knowledge, values, attitudes, commitment, and skills needed to protect and improve the environment
2) To develop conscious towards a cleaner and better managed environment

Course content

1 Introduction - Environmental studies Definition, scope importance and need for public awareness. (Multidisciplinary nature of environmental studies)

2 Environmental Pollution -Definition, Causes, effects on human, water, soil, air (Mother Earth)

- Air pollution
- Water pollution
- Soil pollution
- Marine pollution
- Noise pollution
- Thermal pollution
- Nuclear hazards

3 Various Government initiatives for conservation of Environment. Controlling measures)

- Solid waste Management: Causes, effects and control measures of urban and industrial wastes.
- Role of an individual in prevention of pollution. Pollution case studies.
- Disaster management: floods, earthquake, cyclone and landslides.

4 Field work Visit / Project Report preparation

•Visit to a local area to document environmental assets - river / forest / grassland / hill / mountain.

• Visit to a local polluted site-Urban/Rural/Industrial/Agricultural

• Effects on plants, insects, birds – As Elements of ecosystem

Evaluation of the course: Continuous evaluation of the student through oral, necessary writing assignments / Quiz and presentations.

Certification: A Course Completion Certificate will be provided by the college to every student who has passed in the continuous evaluation and the Grade as per his / her performance in the evaluation will appear on the Certificate.

OR – (Select Any One Course In Semester III – For BBA , BBA- IB , and BBA –CA)

SavitribaiPhule Pune University
Syllabus for BBA (CA) (CBCS 2019 Pattern)
Details for Skill Enhancement (Add-On) Courses
AECC - Course Title: - (N)Advance Course in Environmental Awareness
Credit -2 & Hours -30

Course Objectives

- Understand current concern about our impact on the environment.
- Recognize the things they do affect the environment.
- Promote green practices at home and at work.
- Describe what is being done and what we all can do to help prevent harm to the environment.

Course Contents

- **Environmental and Ecosystem Management:**

Concept and scope, Systems of approaches, Standards – International and National, Ecomark, Environmental accounting and auditing, Green funding and taxes, Trade and environmental management. Ecosystem analysis, Modelling, Monitoring and Planning, Ecotourism and Heritage management, Eco restoration,

- **Management of solid waste**

Different types of solid wastes, Methods of disposal and management of Municipal and thermal power plant generated solid wastes, Bio medical wastes and Hazordous wastes, Recycling of wastes, Power generation and waste minimization techniques.

Sanction and enforcement bodies of environmental laws in India.

Legal, administrative and constitutional provisions for environmental protection in India; Role of Supreme Court and Green Bench of High Court; Public awareness and Government measures; Role of Pressure Groups and NGOs; Concepts and Aspects of Public Interest Litigation (PIL); Public Interest Litigation in India on different Environmental Issues.

- **National and Regional Environmental Issues Resource and its conservation;**

Ecological refugees; Conservation strategies of the environment: Mines, riverine networks; forest, soil and wild life

Current Environmental Movements in India. Silent Valley, Chipko, Narmada dam, Appiko, TehriGarwal Dam, Uttara Kannada and Almatti dam movements.

- **Environmental Ethics and Global Imperatives.**

Concepts and aspects of Environmental ethics, Anthropocentrism and Eco-centrism; Deep ecology.Global environmental problems.Green house effect, global warming and climate change, ozone layer depletion, acid rain, deforestation and loss of biodiversity, unplanned urbanization.

Evaluation of the course: Continuous evaluation of the student through oral, necessary writing assignments/ Quiz and presentations.

Certification: A Course Completion Certificate will be provided by the college to every student who has passed in the continuous evaluation and the Grade as per his / her performance in the evaluation will appear on the Certificate.

S.Y.B.B.A.(C.A.) Semester –IV

Course Code: CA-401

Subject: Networking

Objectives:

1. To gain knowledge about Computer Networks concepts.
2. To know about working of networking models, addresses, transmission medias and connectivity devices.
3. To acquire information about network security and cryptography.

Unit	Topic	No. of Lectures
1	**Introduction to Computer Network** 1.1Basics of Computer Network 1.1.1Definition 1.1.2Goals 1.1.3Applications, 1.1.4Network Hardware –Broadcast, Point to Point 1.1.5Components of Data Communication 1.2 Network Topologies 1.2.1Mesh 1.2.2 Star, 1.2.3 Bus, 1.2.4Ring 1.3Types of Networks 1.3.1LAN,MAN,WAN, 1.3.2 Internetwork, 1.3.3 Wireless Network 1.4 Modes of Communication 1.4.1 Simplex, 1.4.2 Half Duplex, 1.4.3 Full Duplex 1.5. Server Based LANs & Peer-to-Peer LANs 1.6. Protocols and Standards 1.7. Network Software 1.7.1 Protocol Hierarchies,Layers, Peers,Interfaces 1.7.2 Design Issues of the Layers 1.7.3 Connection Oriented and Connectionless Service	**10**
2	**Network Models** 2.1 OSI Reference Model : Functions of each Layer 2.2 TCP/IP Reference Model,Comparison of OSI and TCP/IP	**8**

	Reference Model 2.3 TCP/IP Protocol Suite 2.4 Addressing 2.4.1Physical Addresses 2.4.2 Logical Addresses 2.4.3Port Addresses, 2.4.4 SpecificAddresses 2.5 IP Addressing 2.5.1 ClassfullAddressing 2.5.2 Classless Addressing	
3	**Transmission Media** 3.1Introduction, Types of Transmission Media 3.2 Guided Media: 3.2.1Twisted Pair Cable- Physical Structure,Categories,Connectors &Applications 3.2.2Coaxial Cable – Physical Structure, Standards, Connectors & Applications 3.2.3Fiber Optic Cable- Physical Structure,Propagation Modes,Connectors & Applications 3.3 Unguided Media: 3.3.1Electromagnetic Spectrum for Wireless Communication 3.3.2Propagation Modes Ground,Sky,Line-of-Sight 3.3.3Wireless Transmission:Radio Waves,Microwaves, Infrared	**8**
4	**Wired and Wireless LAN** 4.1 IEEE Standards 4.2 Standard Ethernet MAC Sublayer,Physical Layer 4.3 Fast Ethernet – Goals, MAC Sublayer,Topology, Implementation 4.4 Gigabit Ethernet – Goals, MAC Sublayer,Topology, Implementation 4.5 Ten-Gigabit Ethernet – Goals, MAC Sublayer, Physical Layer 4.6 Backbone Networks -Bus Backbone, Star Backbone 4.7 Virtual LANs Membership, IEEE standards advantages 4.8 Wireless LAN 4.8.1 IEEE 802.11 Architecture, 4.8.2 Bluetooth Architecture (Piconet, Scatternet)	**8**
5	**Network Devices** 5.1 Network Connectivity Devices 5.1.1 Active and Passive Hubs 5.1.2 Repeaters 5.1.3 Bridges- Types of Bridges 5.1.4 Switches 5.1.5 Router 5.1.6 Gateways	**6**

6	**Network Security** 6.1 Introduction 6.2 Need for Security 6.3 Security Services : 6.3.1 Message- -Confidentiality, Integrity, Authentication, Non repudiation. 6.3.2 Entity (User)- Authentication. 6.4 Types of Attack 6.5 Cryptography, PlainText,Cipher Text, Encryption,Decryption, Symmetric Key and Asymmetric Key Cryptography 6.6 SubstitutionTechniques, Caesar Cipher,and Transposition Cipher (Problems should be covered.) 6.7 Firewalls- Packet Filter firewall, Proxy firewall 6.8 Steganography,Copyright	**8**
Total		**48**

Reference Books:

1. Computer Networks by Andrew Tanenbaum, Pearson Education.[4th Edition]
2. Data Communication and Networking by BehrouzForouzan, TATA McGraw Hill. .[4th Edition]

S.Y.B.B.A.(C.A.) Semester –IV

Course Code: CA-402

Subject: Object Oriented Concepts Through CPP

Objectives:

1. Acquire an understanding of basic object-oriented concepts and the issues involved in effective class design.
2. Enable students to write programs using C++ features like operator overloading, constructor and destructor, inheritance, polymorphism and exception handling.

Unit	Topic	No. of Lectures
1	**Introduction to C++** 1.1 Basic concepts, features, advantages and applications of OOP 1.2 Introduction, applications and features of C++ 1.3 Input and Output operator in C++ 1.4 Simple C++ program	2
2	**Beginning with C++** 2.1 Data type and Keywords 2.2 Declaration of variables, dynamic initialization of variables, reference variable 2.3 Operators: 2.3.1 Scope resolution operator 2.3.2 Memory management operators 2.4 Manipulators 2.5 Functions: 2.5.1 Function prototyping, call by reference and return by reference 2.5.2 Inline functions 2.6 Default arguments	6
3	**Classes and Objects** 3.1 Structure and class, Class, Object 3.2 Access specifiers, defining data member 3.3 Defining member functions inside and outside class definition. 3.4 Simple C++ program using class 3.5 Memory allocation for objects 3.6 Static data members and static member functions 3.7 Array of objects, objects as a function argument 3.8 Friend function and Friend class 3.9 Function returning objects	8
4	**Constructors and Destructors** 4.1 Constructors 4.2 Types of constructor : Default, Parameterized, Copy 4.3 Multiple constructors in a class 4.4 Constructors with default argument	6

	4.5 Dynamic initialization of constructor 4.6 Dynamic constructor 4.7 Destructor	
6	**Inheritance** 6.1 Introduction 6.2 Defining Base class and Derived class 6.3 Types of Inheritance 6.4 Virtual Base Class 6.5 Abstract class 6.6 Constructors in derived class	**6**
7	**Polymorphism** 7.1 Compile TimePolymorphism 7.1.1 Introduction, rules for overloading operators 7.1.2 Functionoverloading 7.1.3 Operator Overloading unary and binary 7.1.4 Operator Overloading using friendfunction 7.1.5 Overloading insertion and extractionoperators 7.1.6 String manipulation using operatoroverloading 7.2 RuntimePolymorphism 7.2.1 this Pointer, pointers to objects, pointer to derived classes 7.2.2 Virtual functions and pure virtualfunctions	**8**
8	**Managing console I/O operations** 8.1 C++ streams and C++ streamclasses 8.2 Unformatted I/O operations 8.3 Formatted console I/Ooperations 8.4 Output formatting usingmanipulators 8.5 User defined manipulators	**3**
9	**Working with Files** 9.1 Stream Classes for File operations 9.2 File operations - Opening, Closing andupdating 9.3 File updating with random access. 9.4 Error handling during Fileoperations 9.5 Command Line arguments	**6**
10	**Templates** 10.1 Introduction 10.2 ClassTemplate and class template with multiple parameters 10.3 FunctionTemplate and function template with multiple parameter 10.4 ExceptionHandlingIntroduction	**3**
Total		**48**

Reference Books:

1) Object Oriented programming with C++ by EBalagurusamy
2) Object Oriented Programming with C++ by RobertLafore
3) The Complete Reference C++ by Herbert Schildt
4)

S.Y.B.B.A.(C.A.) Semester-IV

Subject: **Operating System**

Course Code:CA-403

Objectives:

1. To know the services provided by Operating System
2. To know the scheduling concept
3. To understand design issues related to memory management and various related algorithms.
4. To understand design issues related to File management and various related algorithms

Unit	Topic	No. of Lectures
1	**Introduction to Operating System** 1.1 What is operating system 1.2 Computer system architecture 1.3 Services provided by OS 1.4 Types of OS 1.5 Operating System Structure – - Simple structure -Layered approach -Micro kernels -Modules 1.6 Virtual Machines – Introduction, Benefits	3
2	**System Structure** 2.1 User operating system Interface 2.2 System Calls– -Process or job control -Device Management - File Management 2.3 System Program 2.4 Operating System Structure	3
3	**Process Management** 3.1 Process Concept – - The process - Process states - Process control block 3.2 Process Scheduling – - Scheduling queues - Schedulers -Context Switch 3.3 Operation on Process – - Process Creation -Process Termination 3.4 Interprocess Communication –	4

	- Shared memory system - Message passing systems.	
4	**CPU Scheduling** 4.1 What is scheduling 4.2 Scheduling Concepts – - CPU- I/O Burst Cycle - CPU Scheduler -Preemptive and Non-preemptive scheduling - Dispatcher 4.3 Scheduling criteria 4.4 Scheduling Algorithms – - FCFS - SJF (Preemptive& non-preemptive) - Priority Scheduling (Preemptive& Non- preemptive) - Round Robin Scheduling - Multilevel Queues - Multilevel Feedback queues	**6**
5	**Process Synchronization** 5.1 Introduction 5.2 Critical section problem 5.3 Semaphores – - Concept - Implementation - Deadlock & Starvation - Types of Semaphores 5.4 Classical Problems of synchronization – -Bounded buffer problem - Readers & writers problem - Dining Philosophers problem	**6**
6	**Deadlock** 6.1 Introduction 6.2 Deadlock Characterization 6.3 Necessary Condition 6.4 Deadlock Handling Technique– -Deadlock Prevention - Deadlock Avoidance – - Safe State - Resource allocation graph algorithm - Bankers algorithm - Deadlock Detection - Recovery from Deadlock – -Process Termination -Resource Preemption	**7**

7	**Memory Management** 7.1.Background – -Basic hardware - Address binding - Logical versus physical address space - Dynamic loading - Dynamic linking and shared libraries 7.2 Swapping 7.3 Contiguous Memory Allocation – - Memory mapping and protection -Memory allocation - Fragmentation 7.4 Paging – - Basic Method - Hardware support - Protection - Shared Pages 7.5 Segmentation – - Basic concept - Hardware 7.6 Virtual Memory Management – - Background - Demand paging - Performance of demand paging - Page replacement – - FIFO - OPT - LRU - Second chance page replacement - MFU - LFU	**8**
8	**File System** 8.1 Introduction & File concepts (file attributes, Operations on files) 8.2 Access methods – - Sequential access - Direct access 8.3 File structure – - Allocation methods - Contiguous allocation - Linked Allocation - Indexed Allocation 8.4 Free Space Management – - Bit Vector - Linked List - Grouping	**7**

	- Counting	
9	**I/O System** 9.1 Introduction 9.2 I/O Hardware 9.3 Application of I/O Interface 9.4 Kernel I/O Subsystem 9.5 Disk Scheduling – - FCFS - Shortest Seek time first - SCAN - C- SCAN - C- Look	**4**
Total		**48**

Reference Books:

1. Operating System Concepts - Siberchatz, Galvin, Gagne (8th Edition).

2. Operating Systems : Principles and Design – Pabitra Pal Choudhary (PHI Learning Private Limited)

S.Y.B.B.A.(C.A.) Semester – IV

Course Code: CA- 404 (Option)

Course Title : Advance PHP

Objectives -:

1. To know & understand concepts of internet programming.
2. Understand how server-side programming works on the web.
3. Understanding How to use PHP Framework (Joomla / Druple)

Unit No	Topic	No. of Lectures
1	**Introduction to Object Oriented Programming in PHP** 1.1 Classes 1.2 Objects 1.3 Introspection 1.4 Serialization 1.5 Inheritance 1.6 Interfaces 1.7 Encapsulation	6
2	**Web Techniques** 2.1 Server information 2.2 Processing forms 2.3 Sticky forms 2.4 Setting response headers	4
3	**XML** 3.1 Introduction XML 3.2 XML document Structure 3.3 PHP and XML 3.4 XML parser 3.5 The document object model 3.6 The simple XML extension 3.7 Changing a value with simple XML	8
4	**Ajax with PHP** 4.1 Understanding java scripts for AJAX 4.2 AJAX web application model 4.3 AJAX –PHP framework 4.4 Performing AJAX validation 4.5 Handling XML data using php and AJAX 4.6 Connecting database using php and AJAX	6

5	**Introduction to Web Services** 5.1 Definition of web services 5.2 Basic operational model of web services, tools and technologies enabling web services 5.3 Benefits and challenges of using web services. 5.4 Web services Architecture and its characteristics 5.5 Core building blocks of web services 5.6 Standards and technologies available for implementing web services 5.7 Web services communication models 5.8 Basic steps of implementing web services.	10
6	**PHP Framework (Joomla / Druple)** **6.1 Introduction to Joomla/Druple** 6.1.1 Introduction 6.1.2 Joomla/Druple features 6.1.3 How joomla/Drupleworks ? 6.1.4 The platformComponents, Modules and Plugins **6.2 Administering Joomla/Druple** 6.2.1 Presentation Administration 6.2.2 Content Administration 6.2.3 System Administration **6.3 Working with Joomla/Druple** 6.3.1 Adding articles 6.3.2 Adding menus to point to content 6.3.3 Installing new templates 6.3.4 Creating templates 6.3.5 Adding a Module and Component 6.3.6 Modifying the existing templates 6.3.7 Creating templates with web editors 6.3.8 Creating real templates	14

Reference Books

- Php: A Beginner's Guide 1st EditionMcGraw-Hill Osborne Media; 1 edition by VikramVaswani
- Murach's PHP and MySQL (2nd Edition)by Joel Murach and Ray Harris
- PHP: The Complete Reference Paperback – 1 Jul 2017by Steven Holzner (Author)
- Building Web Services with Java, 2nd Edition, S. Graham and others, Pearson Edn., 2008.
- Java Web Services, D.A. Chappell & T. Jewell, O'Reilly,SPD.
- www.php.net.in
- www.W3schools.com

S.Y.B.B.A.(C.A.) Semester – IV

Course Code: CA- 404(Option)

Course Title : Node - JS

Objectives:

1. Understand the JavaScript and technical concepts behind Node JS
2. Structure a Node application in modules
3. Understand and use the Event Emitter
4. Understand Buffers, Streams, and Pipes
5. Build a Web Server in Node and understand how it really works
6. Connect to a SQL or Mongo database in Node

Pre-requisite / Target Audience:

1) Basic Knowledge of JavaScript and OOPS
2) Knowledge in async programming will be added advantage

Unit	Topics	No. of Lectures
1	**Introduction to Node JS** 1.1 Introduction 1.2 What is Node JS? 1.3 Advantages of Node JS 1.4 Traditional Web Server Model 1.5 Node.js Process Model 1.6 Install Node.js on Windows 1.7 Working in REPL	**8**
2	**Node JS Modules** 2.1Functions 2.2 Buffer 2.3 Module 2.4 Module Types 2.5 Core Modules 2.6 Local Modules 2.7 Module.Exports	**10**
3	**Node Package Manager** 3.1 What is NPM ? 3.2 Installing Packages Locally 3.3 Adding dependency in package.json 3.4 Installing packages globally 3.5 Updating packages	**6**
4	**Web server**	

	4.1 Creating web server 4.2 Handling http requests 4.3 Sending requests	**6**
5	**File System** 5.1 Fs.readFile 5.2 Writing a File 5.3 Writing a file asynchronously 5.4 Opening a file 5.5 Deleting a file 5.6 Other IO Operations	**8**
6	**Events** 6.1 EventEmitter class 6.2 Returning event emitter 6.3 Inhering events	**4**
7	**Database connectivity** 7.1 Connection string 7.2 Configuring 7.3 Working with select command 7.4 Updating records 7.5 Deleting records	**6**
Total		**48**

Reference Books:

1) Node.js complete reference guid , velentinBojinov, David Herron, DiogeResende, packt Publishing ltd
2) Mastering Nod.js By SandroPasquali , packt Publishing
3) Smashing Node.js Javascript Everywhere , Guillermo Rauch, John wiley& Sons

Acknowledgement

The Syllabus Restructuring of BBA (CA) Programme (CBCS-2019 Pattern) is a manifestation of excellence in the faculty of Commerce and Management. Savitribai Phule Pune University's focus has always been in raising the academic standards and excellence in the field of education.

The BBA (CA) Programme predominantly endeavours for holistic development of students. It has emphasized on cultivating various skills and has also desired software technology acumen amongst the students.

This revision has been possible only with the help and support of different eminent personalities. The contribution of all the members as a team has enabled the robust revision of all the titles of the Programme. This synergy of the contributors is very crucial in fine tuning of the BBA(CA) Programme in its present form.

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BBA(CA) CBCS 2019 Pattern

TYBBA (CA) Sem V Syllabus

Savitribai Phule Pune University
T.Y.B.B.A (C.A.) Semester –V
Course Code: CA-501
Subject Name: Cyber Security

Total Hours : 48 lectures **Total Credits: 03**

Prerequisites: -

- A course on Computer Networks.

Course Objectives:

- To understand the fundamentals of cyber security.
- To understand various categories of Cybercrime, Cyber-attacks on mobile, tools and techniques used in Cybercrime and case studies.
- To have an overview of the Cyber laws and concepts of Cyber forensics.

Course Outcome:-

- Have a good understanding of Cyber Security and the Tools.
- Identify the different types of Cyber Crimes.
- Have a good understanding of Cyber laws
- To develop Cyber forensics awareness.
- Identify attacks, security policies and credit card frauds in mobile and Wireless Computing Era.

Unit	Topic	No of lectures
1	**Chapter 1:- Introduction to Cyber Crime and Cyber Security** 1.1 Introduction 1.2 Cybercrime: Definition and Origin of the Word 1.3 Cybercrime and Information Security 1.4 Who are Cybercriminals? 1.5 Classifications of Cybercrimes: E-Mail Spoofing, Spamming, Cyber defamation, Internet Time Theft, Salami Attack/Salami Technique, Data Diddling,Forgery, Web Jacking, Newsgroup, Spam/Crimes Emanating from Usenet Newsgroup, Industrial Spying/Industrial Espionage, Hacking,OnlineFrauds,Computer Sabotage, Email Bombing/Mail Bombs, Computer Network Intrusions, Password Sniffing, Credit Card Frauds, Identity Theft 1.6 Definition of Cyber Security 1.7 Vulnerability, Threats and Harmful acts 1.8 CIA Triad 1.9 Cyber Security Policy and Domains of Cyber Security Policy	07
2	**Chapter 2 :- Cyber offenses and Cyberstalking** 2.1 Criminals Plan: Categories of Cybercrime Cyber Attacks: Reconnaissance, Passive Attack, Active Attacks, Scanning/Scrutinizing gathered Information, Attack (Gaining and Maintaining the System Access), Social Engineering, and Classification of Social Engineering. 2.2 Cyberstalking: Types of Stalkers, Cases Reported on Cyberstalking, Working of Stalking 2.3 Real-Life Incident of Cyber stalking 2.4 Cybercafe and Cybercrimes	10

	2.5 Botnets: The Fuel for Cybercrime, Botnet, Attack Vector 2.6 Cybercrime: Mobile and Wireless Devices – Proliferation - Trends in Mobility 2.7 Credit Card Frauds in Mobile and Wireless Computing Era 2.8 Security Challenges Posed by Mobile Devices 2.9 Authentication Service Security 2.10 Attacks on Mobile/Cell Phones	
3	**Chapter 3:- Tools and Methods Used in Cybercrime** 3.1 Introduction 3.2 Proxy Servers and Anonymizers 3.3 Phishing 3.4 Password Cracking 3.5 Keyloggers and Spywares 3.6 Virus and Worms 3.7 Trojan Horses and Backdoors 3.8 Steganography 3.9 DoS and DDoS Attacks 3.10 SQL Injection	05
4	**Chapter 4 :- Cybercrimes and Cyber security: The Legal Perspectives** 4.1 Introduction 4.2 Cybercrime and the Legal Landscape around the World 4.3 Why Do We Need Cyberlaws: The Indian Context 4.4 The Indian IT Act 4.5 Challenges to Indian Law and Cybercrime Scenario in India 4.6 Consequences of not Addressing the Weakness in Information Technology Act 4.7 Digital Signatures and the Indian IT Act 4.8 Amendments to the Indian IT Act 4.9 Cybercrime and Punishment 4.10 Cyberlaw, Technology and Students: Indian Scenario	07
5	**Chapter 5:- Cyber Forensics** 5.1 Introduction 5.2 Historical background of Cyber forensics 5.3 Digital Forensics Science 5.4 The Need for Computer Forensics 5.5 Cyber Forensics and Digital evidence 5.6 Forensics Analysis of Email 5.7 Digital Forensics Lifecycle 5.8 Challenges in Computer Forensics	06
6	**Chapter 6:- Cybersecurity: Organizational Implications** 6.1 Organizational Implications: Cost of cybercrimes and IPR issues 6.2 Web threats for organizations 6.3 Security and Privacy Implications from Cloud Computing 6.4 Social media marketing 6.5 Social computing and the associated challenges for organizations, Protecting people's privacy in the organization 6.6 Organizational guidelines for Internet usage and safe computing guidelines and computer usage policy 6.7 Incident handling	07

	6.8 Intellectual property in the cyberspace of cyber security.	
7	**Chapter 7:- Cybercrime: Illustrations, Examples and Mini-Cases** 7.1Real-Life Examples 7.2 Mini-Cases 7.3 Illustrations of Financial Frauds in Cyber Domain 7.4 Digital Signature-Related Crime Scenarios 7.5 Digital Forensics Case Illustrations 7.6 Online Scams	06

References Books:

1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives – Nina Godbole, SunitBelapure, Wiley: April 2011 India Publications Released.
2. Principles of Information Security,-Michael E Whitman, Herbert J Mattord, 3rd Edition, 2011.
3. Computer Security: Principles and Practice -William Stallings and Lawrie Brown, 3rd edition, Pearson, 2015.
4. Cyber Security Essentials- James Graham Richard Howard Ryan Olson

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Semester –V
Course Code: CA-502
Subject: Object Oriented Software Engineering

Total Hours: 48 **Total Credits: 03**

Pre Requisite: Students shall have the Basic Knowledge of Software Engineering

OBJECTIVES:

1. To understand the fundamentals of object modeling
2. To understand and differentiate Unified Process from other approaches.
3. To design with static UML diagrams.
4. To design with the UML dynamic and implementation diagrams.
5. To improve the software design with design patterns.
6. To test the software against its requirements specification.

Outcomes:

1. Students will be able to give Design Specifications for Project.
2. Students will acquire Knowledge in Basic Modeling.
3. Students will acquire Project Management Skills.

Chapter	Course Content	No of lectures
1	**Introduction and basics of Software Modelling** 1.1 Software Life Cycle Models (Revision of SE) 1.2 System Concepts 1.3 Project Organization 1.4 Communication in Project Management 1.5 Risk management in Project Management	4
2	**SRS Documentation** 2.1 SRS Specification 2.2 Requirement Elicitation **2.3** Business Engineering	4
3	**Introduction to UML** 3.1 Concept of UML 3.2 Advantages of UML	2
4	**Object Oriented Concepts and Principles** 4.1 What is Object Orientation? - Introduction , Object , Classes and Instance , Polymorphism, Inheritance 4.2 Object Oriented System Development- Introduction, Function/Data Methods (With Visibility), Object Oriented Analysis, Object Oriented Construction 4.3 Identifying the Elements of an Object Model 4.4 Identifying Classes and Objects 4.5 Specifying the Attributes (With Visibility)	4

	4.6 Defining Operations 4.7 Finalizing the Object Definition	
5	**Structural Modeling** 5.1 Classes 5.2 Relationship 5.3 Common Mechanism 5.4 Class Diagram (Minimum three examples should be covered) 5.5 Advanced Classes 5.6 Advanced Relationship 5.7 Interface 5.8 Types and Roles 5.9 Packages 5.10 Object Diagram (Minimum three examples should be covered)	**10**
6	**Basic Behavioural Modeling** 6.1 Interactions 6.2 Use Cases and Use Case Diagram with stereo types (Minimum three examples should be covered) 6.3 Interaction Diagram (Minimum two examples should be covered) 6.4 Sequence Diagram (Minimum two examples should be covered) 6.5 Activity Diagram (Minimum two examples should be covered) 6.6 State Chart Diagram (Minimum two examples should be covered)	**10**
7	**Architectural Modelling** 7.1 Component 7.2 Components Diagram (Minimum two examples should be covered) 7.3 Deployment Diagram (Minimum two examples should be covered) 7.4 Collaboration Diagram (Minimum two examples should be covered)	**6**
8	**Object Oriented Analysis** 8.1 Iterative Development and the Rational Unified Process 8.2 Inception 8.3 Understanding Requirements 8.4 Use Case Model From Inception to Elaboration 8.5 Elaboration	**4**
9	**Object Oriented Design** 9.1 The Booch Method, The Coad and Yourdon Method and Jacobson Method and Raumbaugh Method 9.2 The Generic Components of the OO Design Model	**4**

	9.3 The System Design Process - Partitioning the Analysis Model, Concurrency and Sub System Allocation, Task Management Component, The Data Management Component, The Resource Management Component, Inter Sub System Communication	
	Total	**48**

Reference Books:

Sr. No.	Title of the Book	Author's Name	Publication
1	The Unified Modeling Language User/Reference Guide,	**Grady Booch, James Rambaugh**	Pearson Education Inc
2	The Unified software development Process	**Ivar Jacobson, Grady Booch, James Rambaugh**	**Pearson Education**
3	**Agile Software development**	**Alistair Cockbair**	**Pearson Education**

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Semester –V
Course Code: CA-503
Subject: Core Java

Total Hours : 48 **Total Credits: 03**

Prerequisite:

- Student should know basics of object oriented programming.

Course Objectives:

- To introduce the object oriented programming concepts.
- To understand object oriented programming concepts, and apply them in solving problems.
- To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes
- To introduce the implementation of packages and interfaces
- To introduce the concepts of exception handling and multithreading.
- To introduce the design of Graphical User Interface using applets and swing controls.

Course Outcomes:

- Able to solve real world problems using OOP techniques.
- Able to understand the use of abstract classes.
- Able to solve problems using java collection framework and I/o classes.
- Able to develop multithreaded applications with synchronization.
- Able to develop applets for web applications.
- Able to design GUI based applications

Unit No.	Topic	No. of Lectures	Reference Books
1	**Java Fundamentals** 1.1 Introduction to Java. 1.1 Features of Java 1.2 Basics of Java: - Data types, variable, expression, operators, constant. 1.3 Structure of Java Program. 1.4 Execution Process of java Program. 1.5 JDK Tools. 1.6 Command Line Arguments. 1.7 Array and String: 1.7.1 Single Array & Multidimensional Array 1.7.2 String, String Buffer 1.8 Built In Packages and Classes : 1.8.1 java.util:- Scanner, Date, Math etc. 1.8.2 java.lang	8	1,2
2	**Classes, Objects and Methods** 2.1 Class and Object 2.2 Object reference 2.3 Constructor: Constructor Overloading 2.4 Method: Method Overloading, Recursion, Passing and Returning object form Method 2.5 new operator, this and static keyword, finalize() method 2.6 Nested class, Inner class, and Anonymous inner class	8	1,2

3	**Inheritance, Package and Collection** 3.1 Overview of Inheritance 3.2 inheritance in constructor 3.3 Inheriting Data members and Methods, 3.4 Multilevel Inheritance – method overriding Handle multilevel constructors 3.5 Use of super and final keyword 3.6 Interface: 3.7 Creation and Implementation of an interface, Interface reference 3.8 Interface inheritance 3.9 Dynamic method dispatch 3.10 Abstract class 3.11 Comparison between Abstract Class and interface **3.12** Access control **3.13 Packages** 3.13.1 Packages Concept 3.13.2 Creating user defined packages 3.13.3 Java Built inpackages 3.13.4 Import statement, Static import 3.14 **Collection** 3.14.1 CollectionFramework. 3.14.2 Interfaces: Collection, List, Set 3.14.3 Navigation: Enumeration, Iterator, ListIterator 3.14.4 Classes: LinkedList, ArrayList, Vector, HashSet	10	
4	**File and Exception Handling** **Exception** 4.1 Exception and Error 4.2 Use of try, catch, throw, throws and finally 4.3 Built in Exception 4.4 Custom exception 4.5 Throwable Class. **File Handling** 4.6 Overview of Different Stream (Byte Stream, Character stream) 4.7 Readers and Writers class 4.8 File Class 4.9 File Input Stream , File Output Stream 4.10 Input Stream Reader and Output Stream Writer class 4.11 FileReader and FileWriter class 4.12 Buffered Reader class.	8	1,2,3
5	**Applet, AWT, Event and Swing Programming** **Applet** 5.1 Introduction 5.2 Typesapplet 5.3 Applet Lifecycle 5.3.1 Creatingapplet 5.3.2 Applet tag	14	1,2,3,4

	5.4 AppletClasses 5.4.1 Color 5.4.2 Graphics 5.4.3 Font **AWT** 5.5 Components and container used inAWT 5.6 Layoutmanagers 5.7 Listeners and Adapterclasses 5.8 Event Delegationmodel **Swing** 5.9 Introduction to Swing Componentand Container Classes 5.10 Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JTable, JComboBox, Swing Menus, Dialogs,JFileOpen,JColorChooser.		
	Total Lectures	**48**	

Reference Books:

1. Programming with JAVA - EBalgurusamy
2. The Complete Reference – JAVA HerbertSchildt
3. Programming in Java, S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
4. Java Programming and Object-oriented Application Development, R. A. Johnson, Ceng

T.Y.B.B.A.(C.A.) Semester –V
Course Code: CA-504
Subject: MongoDB

Total Hours: 48 **Total Creidts:03**

Prerequisites:

- Knowledge of database concepts
- Basic understanding of Big Data platforms

Objectives:

1. Understand importance of NoSQL Databases.
2. Learn various MongoDB commands and MongoDB design goals.
3. Design basic and general-purpose database using MongoDB.

Outcomes:

- Learned to work with MongoDB shell and MongoDB tools.
- Able to do Schema design, Data modelling and all sorts of CRUD Operations.
- Learned to optimize query performance.
- Become capable to analyze the data stored in MongoDB.

Unit	Topic	No. of lectures
1	**Introduction to NoSQL Databases** 1.1 Introduction to NoSQL Databases 1.2 Difference between NoSQL and RDBMS 1.3 Need of NoSQL Databases 1.4 Application of NoSQL Databases 1.5 Types of NoSQL Databases 1.6 What is MongoDB? 1.7 Features of MongoDB	5
2	**MongoDB Basics** 2.1 Installing MongoDB 2.2 MongoDB Server and Database, MongoDB tools 2.3 Collection, Documents and Key-Values 2.4 Data Modeling Concepts 2.4.1 Why Data Modeling? Data Modeling Approach 2.4.2 Analogy between RDBMS & MongoDB Data Model, MongoDB Data 2.4.3 Model (Embedding & Linking) 2.4.4 Challenges for Data Modeling in MongoDB 2.4.5 Data Model Examples and Patterns 2.5 Mongo shell Commands to create, delete database, collection & documents 2.6 MongoDB Datatypes 2.7 Inserting and saving documents 2.7.1 Batch Insert 2.7.2 Insert Validation 2.8 MongoDB GUI like compass	12
3	**MongoDB CRUD Operations**	14

	3.1 MongoDB Development Architecture 3.2 MongoDB Production Architecture 3.3 MongoDB CRUD Introduction, MongoDB CRUD Concepts 3.4 MongoDB CRUD Concerns (Read & Write Operations) 3.5 Concern Levels, Journaling 3.6 Cursor Query Optimizations, Query behaviour in MongoDB 3.7 Distributed Read & Write Queries 3.8 MongoDB CRUD Syntax & Queries	
4	**MongoDB Index and Aggregation** 4.1 Index Introduction, Index Concepts, Index Types, Index Properties 4.2 Index Creation and Indexing Reference 4.3 Introduction to Aggregation 4.4 Approach to Aggregation 4.5 Types of Aggregation (Pipeline, MapReduce & Single Purpose) 4.6 Performance Tuning.	**8**
5	MongoDB Administration 5.1 Administration concepts in MongoDB 5.2 Monitoring issues related to Database 5.3 Monitoring at Server, Database, Collection level, and various Monitoring tools related to MongoDB 5.4 Database Profiling, Locks, Memory Usage, No of connections, page fault 5.5 Backup and Recovery Methods for MongoDB 5.6 Export and Import of Data to and from MongoDB 5.7 Run time configuration of MongoDB 5.8 Production notes/ best practices 5.9 Data Managements in MongoDB (Capped Collections/ Expired data from TTL), Hands on Administrative Tasks.	**9**
Total		**48**

Reference books:

1. MongoDB Basicsby Peter Membrey, David Hows, EelcoPlugge
2. MongoDB RecipesWith Data Modeling and Query Building StrategiesbySubhashiniChellappan, DharanitharanGanesan
3. MongoDB Simply In DepthbyAjit Singh, Sultan Ahmad

Subject Code: 504
Subject: Python

Total Hours :- 48 **Total Credits: 03**

Prerequisites:

1. Experience with a high level language (C/C++, Java, MATLAB) is suggested.
2. Prior knowledge of a scriptinglanguage (Perl, UNIX/Linux shells) and Object-Oriented concepts is helpful but not mandatory.

Course Objectives:

1. To learn and understand Python programming basics and paradigm.
2. To learn and understand python looping, control statements and string manipulations.
3. Students should be made familiar with the concepts of GUI controls and designing GUI applications.
4. To learn and know the concepts of file handling, exception handling.

Course Outcomes: On completion of the course, student will be able

1. Define and demonstrate the use of built-in data structures "lists" and "dictionary".
2. Design and implement a program to solve a real world problem.
3. Design and implement GUI application and how to handle exceptions and files.

Unit	Details	Lectures
I	**Unit 1: Introduction to Python** 1.1 History, feature of Python, setting up path, working with python Interpreter, basic syntax, variable and data types, operators 1.2 **Conditional statements**-If, If-Else, nested if-else, Examples. 1.3 **Looping-**For,While,Nested loops, Examples 1.4 **Control Statements**-Break, Continue, Pass. 1.5 **String Manipulation**-Accessing String, Basic Operations, String Slices, Function and Methods, Examples. 1.6 **Lists**-Introduction, accessing list, operations, working with lists, function & methods. 1.7 **Tuple**-Introduction, Accessing tuples, operations working, function & methods, Examples. 1.8 **Dictionaries**-Introduction, Accessing values in dictionaries, working with dictionaries, properties, function, Examples. 1.9 **Functions**-Defining a function, calling a function, types of function, function arguments, anonymous function, global & local variable, Examples.	**16**
II	**Unit 2: Modules and Packages** **2.1Built in Modules** 2.1.1 Importing modules in python program 2.1.2 Working with Random Modules. 2.1.3 E.g. - built-ins, time, date time, calendar, sys, etc **2.2 User Defined functions** 2.2.1Structure of Python Modules **2.3 Packages** 2.3.1 Predefined Packages 2.3.2User defined Packages	**6**
III	**Unit 3: Classes ,Objects and Inheritance** **3.1 Classes and Objects** 3.1.1 Classes as User Defined Data Type 3.1.2 Objects as Instances of Classes 3.1.3 Creating Class and Objects 3.1.4 Creating Objects By Passing Values 3.1.5 Variables & Methods in a Class **3.2 Inheritance** 3.2.1 Single Inheritance 3.2.2 Multilevel Inheritance	**8**

	3.2.3 Multiple Inheritance 3.2.4 Hybrid Inheritance 3.2.5 Hierarchical Inheritance 3.2.6 IS-A Relationship and HAS-A Relationship	
IV	**Unit 4: Exception Handling** 4.1 Python Exception 4.2 Common Exception 4.3 Exception handling in Python (try-except-else) 4.4 The except statement with no exception 4.5 Multiple Exception 4.6 The try-finally clause 4.7 Custom Exception and assert statement	**4**
V	**Unit 5: GUI Programming** 5.1 Introduction 5.2 Tkinter programming 5.4 Tkinter widgets 5.5 Frame 5.6 Button 5.7 Label 5.8 Entry	**10**
VI	**Unit 6: Python Libraries** 6.1 Statistical Analysis- NumPy, SciPy, Pandas, StatsModels 6.2 Data Visualization- Matplotlib, Seaborn, Plotly 6.3 Data Modelling and Machine Learning- Scikit-learn, XGBoost, Eli5 6.4 Deep Learning- TensorFlow, Pytorch, Keras 6.5 Natural Language Processing (NLP)- NLTK, SpaCy, Gensim	**4**

Reference Books:

1.Mark Lutz, Programming Python, O`Reilly, 4th Edition, 2010
2.Dive into Python, Mike
3. Learning Python, 4th Edition by Mark Lutz
4. Programming Python, 4th Edition by Mark Lutz
5.Python Programming:An introduction to computer,John Zelle,3rd Edition.

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Sem-V
Subject Code: 505

Subject: (DSE) Project

Total Credits: 04

For the evaluation/ conduction of project separate guidelines will be provided.

T.Y.B.B.A.(C.A.) Sem-V
Subject Code: 506
Subject: Computer Laboratory Based on 503 and 504(2 credits each)

Total Credits: 04

For the conduction of practicals, practical assignments are given in the lab book.

Savitribai Phule Pune University

T.Y.B.B.A.(C.A.)Sem-V (CBCS 2019 Pattern)
Subject Code: CA-507
Subject: Internet of Things (IoT)

Total Hours: 30 **Total Credits: 02**

Prerequisite:

Basic knowledge of Internet, Networking, and Electronics.

Course Objectives:

1. To understand Technical aspects of Internet of things.
2. To describe smart objects and IoT Architecture.
3. To study and compare different Application protocols ofIoT.
4. To understand IoT platform using Arduino Uno.

Course Outcomes: Students will be able

1. To explain key technologies, smart objects, IoT Architecture and security in Internet of Things.
2. To illustrate the role of IoT protocols for efficient network communication.
3. To understand IoT platform such as Arduino Uno.

Unit No.	Contents Theory	No. of Lectures
1	**Fundamentals of IoT** 1.1 Basic Concepts of IoT 1.2 Major components of IoT devices 1.3 IOT Architecture 1.4 Pros & Cons of IOT	03
2	**Communication Technologies** 2.1 Wireless Communication: Bluetooth, ZigBee, WiFi, RF Links 2.2 Wired Communication: Ethernet 2.3 IOT Protocol: MQTT, CoAP, XMPP, OSGi	05
3	**Microcontroller Fundamental and Arduino uno** 3.1 System on Chip & Microcontroller 3.2 Arduino UNO: Introduction to Arduino, Arduino UNO, Arduino Board, The Anatomy of an Arduino Board 3.3 The Development Environment of Arduino Board 3.4 Writing Arduino Software, The Arduino Sketch 3.5 Fundamentals of Arduino Programming 3.6 Trying the code on an Arduino Emulator 3.7 Arduino Libraries 25 Programming & Interfacing 3.8 Application of IoT 3.9 Case studies: Home Automation, Smart Parking, etc.	07
Total		**15**
Practical Please Refer Lab Book		**15**

Reference Books:

1. Learning internet of things by Waher, Peter -Packt Publishing Ltd, 2015
2. "Fundamentals of Wireless Sensor Networks: Theory and Practice" by WaltenegusDargie,

Christian Poellabauer
3. Internet of Things (A Hands-on-Approach) by Vijay Madisetti , ArshdeepBahga
4. Designing the Internet of Things by Adrian McEwen, Hakim Cassimally
5. Internet of Things with Arduino Cookbook by Schwartz, M. - Packt Publishing Ltd.
6. "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, 1st Edition, Pearson Education (Cisco Press Indian Reprint)
7. "Internet of Things" by Srinivasa K G, CENGAGE Leaning India, 2017
8. Computer Networks by Tanenbaum, Andrew S - Pearson Education Pte. Ltd., Delhi, 4th Edition
9. Data and Computer Communications; By: Stallings, William - Pearson Education Pte. Ltd., Delhi, 6th Edition

TYBBA (CA) Sem VI Syllabus

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Sem-VI (CBCS 2019 Pattern)

Subject Code: CA-601

Subject: Recent Trends in IT

Total Hours: 48 **Total Credits: 3+1=4**

Prerequisites:

1. Basic knowledge of related programming and database concepts.

2. Fundamentals of Mathematical logic & Data structures.

Course Objectives

1. To introduce upcoming trends in Information technology.
2. To study Eco friendly software development concepts.
3. To provide a strong foundation of fundamental concepts in Artificial Intelligence.
4. To evaluate the performance of various data mining task.
5. To understand Data analytics using Spark Programming.

Course Outcomes: On completion of the course, student will be able

1. To discuss the basic concepts AI.
2. To apply basic, intermediate and advanced techniques to mine the data.
3. To provide an overview of the concept of Spark programming.

Unit No.	Contents	No. of Lectures
1	**Introduction to recent trends** 1.1 Artificial Intelligence 1.2 Data Warehouse 1.3 Data Mining 1.4 Spark	02
2	**Artificial Intelligence** 2.1 Introduction& Concept of AI 2.2 Applications of AI 2.3 Artificial Intelligence, Intelligent Systems, Knowledge –based Systems, AI Techniques 2.4 Early work in AI & related fields. 2.5 Defining AI problems as a State Space Search 2.6 Search and Control Strategies 2.7 Problem Characteristics 2.8 AI Problem: Water Jug Problem, Tower of Hanoi, Missionaries & Cannibal Problem	08
3	**AI Search Techniques** 3.1 Blind Search Techniques: BFS, DFS, DLS, Iterative deepening Search, Bidirectional Search, and Uniform cost Search 3.2 Heuristic search techniques: Generate and test, Hill Climbing, Best First search, Constraint Satisfaction, Mean-End Analysis, A*, AO*	08
4	**Data Warehousing** 4.1 Introduction to Data warehouse 4.2 Structure of Data Warehouse 4.3 Advantages & uses of Data Warehouse 4.4 Architecture of Data Warehouse 4.5 Multidimensional data model	08

	4.6 OLAP Vs. OLTP 4.7 OLAP Operations 4.8 Types of OLAP Servers: ROLAP versus MOLAP versus HOLAP	
5	**Data Mining** 5.1 Introduction to Data Mining 5.2 Data mining Task 5.3 Data mining issues 5.4 Data Mining versus Knowledge Discovery in Databases 5.5 Data Mining Verification vs. Discovery 5.6 Data Pre-processing – Need, Data Cleaning, Data Integration & Transformation, Data Reduction 5.7 Accuracy Measures: Precision, recall, F-measure, confusion matrix, cross-validation, bootstrap 5.8 Data Mining Techniques 5.9 Frequent item-sets and Association rule mining: Apriori algorithm, FP tree algorithm 5.10 Graph Mining: Frequent sub-graph mining 5.11 Software for data mining : R, Weka, Sample applications of data mining 5.12 Introduction to Text Mining, Web Mining, Spatial Mining, Temporal Mining	**12**
6	**Spark** 6.1 Introduction to Apache Spark 6.2 Spark Installation 6.3 Apache Spark Architecture 6.4 Components of Spark 6.5 Spark RDDs 6.6 RDD Operations: Transformation & Actions 6.7 Spark SQL and Data Frames 6.8 Introduction to Kafka for Spark Streaming	**10**
Total		**48**

Reference Books:

1. Artificial Intelligence by Elaine Rich, Kevin Knight - Tata McGraw Hill, 2nd Edition
2. Artificial Intelligence: A new Synthesis, Nilsson, Elsevier,ISBN 9788181471901
3. Data Mining Concepts and Techniques, by Jiawei Micheline Kamber, Morgan Kauf Mann Publishers.
4. Advanced Analytics with Spark by Sandy RyzaPublicatio O'REILLY
5. Apache Spark for Data Science Cookbook by Padma Priya Chitturi

Savitribai Phule Pune University

T.Y.B.B.A.(C.A.) Sem-VI (CBCS 2019 Pattern)
Subject Code: CA-602
Subject: Software Testing

Total Hours: 48 **Total Credits: 3**

Prerequisite:

1. Students shall have basic Knowledge of Software Engineering.
2. Students shall have basic Knowledge of OOSE.

Objectives:

1. To provide learner with knowledge in Software Testing techniques.
2. To understand how testing methods can be used as an effective tool in providing quality assurance for software.
3. To provide skills to design test case plan for testing software.

Outcomes:

1. Students will be introduced to testing tools.
2. Students will acquire Knowledge of Basic SQA.
3. Students will be able to design basic Test Cases.

Chapter	Course Content	No of lectures
1	**Introduction** 1.1 Introduction, Nature of errors, 1.2 Testing Objectives 1.3 Testing principles 1.4 Testing fundamentals, 1.5 Software reviews, Formal Technical reviews, 1.6 Inspection and walkthrough 1.7 Testing Life Cycle	**10**
2	**Approaches to Testing –Testing Methods** 2.1 White Box Testing and types of white box testing 2.2 Test Case Design 2.3 Black Box Testing and types of black box testing 2.4 Gray Box Testing	**5**
3	**Software Testing Strategies &Software metrics** 3.1 Software Testing Process 3.2 Unit Testing 3.3 Integration- Top-down ,Bottom up 3.4 System Testing 3.5 Acceptance Testing (alpha, Beta testing) 3.6 Validation and Verification 3.7 Big Bang Approach 3.8 Sandwich approach 3.9 Performance Testing 3.10 Regression Testing 3.11 Smoke Testing 3.13 Load Testing	**10**
4	**Software metrics** 4.1 Introduction 4.2 Basic Metrics –size-oriented metric, Function –oriented metric 4.3 Cyclometic Complexity Metrics Examples on Cyclometic Complexity	**10**
5	**Testing for Specialized Environments** 5.1 Testing GUI's 5.2 Testing of Client/Server Architectures 5.3 Testing Documentation and Help Facilities 5.4 Testing for Real-Time Systems	**5**

6	**Testing Tools& Software Quality Assurance (Introduction)** 6.1 JUnit, Apache JMeter, Win runner 6.2 Load runner, Rational Robot 6.3 Quality Concepts, Quality Movement, Background Issues, SQA activities 6.4 Formal approaches to SQA 6.5 Statistical Quality Assurance 6.6 Software Reliability 6.7 The ISO 9000 Quality Standards 6.8 SQA Plan 6.9 Six sigma 6.10 Informal Reviews	**8**
	Total	**48**

Reference Books:

Sr. No.	Title of the Book	Author's Name	Publication
1.	**Software Engineering – A Practitioner's approach**	**Roger S Pressman**	**7th Edition Tata McGraw-Hill**
2.	Effective Methods of Software Testing.	**William E Perry**	Wiley Publishing Inc
3.	**Software Testing Principles and Practices**	**Srinivasan Desikan, Gopalswamy Ramesh**	**Pearson Publication**
4.	**Total Quality Management**	**DaleH. Besterfield,**	**Prentice Hall, 2003**

Savitribai Phule Pune University

T.Y.B.B.A.(C.A.) Sem-VI (CBCS 2019 Pattern)
Subject Code: CA-603
Subject: Advanced Java

Total Hours: 48 **Total Credits: 3**

Prerequisite: Students should know basic programming concepts.

Objectives -:

1. To know the concept of Java Programming.
2. To understand how to use programming in day to day applications.

3. To develop programming logic.

Outcomes:

1. Students will know the concepts of JDBC Programming.
2. Students will know the concepts of Multithreading and Socket Programming.
3. Students will know the concepts of Spring and Hibernate.
4. Students will develop the project by using JSP and JDBC.
5. Students will develop applications in Spring and hibernate.

Sr. No	**Topic**	Number Of Lectures
1.	**JDBC** 1.1 Introduction 1.2 JDBC Architecture. 1.3 JDBC Process 1.4 **Working with ResultSet Interface.**	**8**
2	**Multithreading:** 2.1 Introduction to Multithreading. 2.2 Thread creation: Thread Class, Runnable Interface. 2.3 Life cycle of Thread. 2.4 Thread Priority. 2.5 Execution of Thread Application. 2.6 Synchronization and Interthread communication.	**12**
3	**Networking:** 3.1 Overview of Networking. 3.2 Networking Basics: Port Number, Protocols and classes. 3.3 Sockets, Reading from and Writing to a Socket.	**5**
4	**Servlet and JSP** 4.1 Introduction to Servlet 4.2 Types of Servlet: Generic Servlet and Http Servlet 4.3 Life cycle of servlet 4.4 Session Tracking. 4.5 Servlet with database. **JSP** 4.6 Introduction to JSP. 4.7 JSP Life Cycle. 4.8 Components of JSP. 4.9 JSP with Database.	**12**
5	**Spring & Hibernate** **Spring:** 5.1 Introduction 5.2 Applications and Benefits of spring 5.3 Architecture and Environment Setup 5.4 Hello World Example 5.5 Core Spring- IoC Containers, Spring Bean Definition, Scope, Lifecycle **Hibernate** 5.6 Architecture and Environment 5.7 Configuration, Sessions, Persistent Class 5.8 Mapping Files, Mapping Types 5.9 Examples	**11**

Reference Books:

1. The Complete Reference – JAVA Herbert Schildt
2. Professional Hibernate, by Eric Pugh, Joseph D. Gradecki by Wiley Publishing, Inc., ISBN: 0- 7645-7677-1
3. Spring In Action, Craig Walls, Ryan Breidenbach, Manning Publishing Co., ISBN: 1-932394- 35-4
4. Head First Servlets and JSP: Passing the Sun Certified Web Component Developer Exam -2nd Edition-Bryan Basham, Kathy Sierra, Bert Bates- O'REILLY.

Savitribai Phule Pune University

T.Y.B.B.A.(C.A.) Sem-VI (CBCS 2019 Pattern)
Subject Code: CA-604
Subject: Android Programming

Total Hours: 48 **Total Credits: 3**

Pre-requisite:

1. Concepts of OOP's.
2. Basic Knowledge About JAVA Programming

Objective:

1. To understand the Android Operating System and develop applications using Google's Android open-source platform.
2. To understand the issues relating to Wireless applications.

Outcome:

1. Student will be able to write simple GUI applications, use built-in widgets and components, work with the database to store data locally, and much more.
2. Demonstrate their understanding of the fundamentals of Android operating systems
   Demonstrate their skills of using Android software development tools

Unit	Topic	No. of lectures
1	**INTRODUCTION TO Android Programming** 1.1 What is Android? 1.2 History and Versions 1.3 Android Architecture 1.4 Basic Building Blocks 1.5 Android API Levels 1.6 Application Structure 1.7 First Hello World Program	**04**
2	**ACTIVITY, INTENT AND LAYOUT** 2.1 Introduction to Activity 2.2 Activity life cycle 2.3 Introduction to Intent 2.4 Types of Intent(Implicit and Explicit Intent) 2.5 Layout Manager 2.5.1View and View Group 2.5.2 Linear Layout 2.5.3 Relative Layout 2.5.4 Table Layout 2.5.5 Grid Layout 2.5.6 Constraint Layout 2.5.7 Frame Layout 2.5.8 Scroll Layout	**07**
3	**BASIC UI DESIGN** 3.1 Button(Push Button, Check Box, Radio Button, Toggle Button, Image Button) 3.2 Text Fields 3.3 Spinner 3.4 List View 3.5 Toast 3.6 Scroll View 3.6 ProgressBar View 3.7 Auto Complete Text View 3.8 Dialog Box 3.8.1 Alert Dialog. 3.8.2 DatePicker Dialog. 3.8.3 TimePicker Dialog. 3.8.4 Custom Dialog.	**10**
4	**ADAPTER AND MENU** 4.1 Base Adapter 4.2 Array Adapter 4.3 ListView using Adapter 4.4GridView using Adapter 4.5Photo Gallery using Adapter	**05**

	4.6 Using Menu with Views 4.6.1 Option Menu 4.5.2 Context Menu 4.5.3 Popup Menu	
5	**THREADS AND NOTIFICATION** 5.1 Worker thread 5.2 Handlers & Runnable 5.3 AsynTask (in detail) 5.4 Broadcast Receiver 5.5 Services 5.5.1Service life Cycle 5.5.2 Bounded Service 5.5.2 Unbounded Service 5.6 Notification 5.7 Alarm 5.8 Accessing Phone services(Call,SMS)	**06**
6	**CONTENT PROVIDER** 6.1Content Providers 6.2 SQLite Programming 6.3 SQLiteOpenHelper 6.4 SQLiteDatabse 6.5 Cursor 6.6 Searching for content 6.7 Adding, changing, and removing content 6.8 Building and executing queries 6.9 Android JSON	**08**
7	**LOCATION BASED SERVICES AND GOOGLE MAP** **7.1** Display Google Maps 7.1.1 Creating the project 7.1.2 Obtaining the Maps API Key 7.1.3 Displaying the Map 7.1.4 Displaying the Zoom Control 7.1.5 Changing Views 7.1.6 Navigating to a specific location 7.1.7 Adding Markers 7.1.8 Getting the location that was touched 7.1.9 Geocoding and Reverse Geocoding 7.2. Getting Location Data 7.3. Monitoring a Location	**08**
	Total Lectures	**48**

Reference Books:

1. Beginning Android4 Application Development, By Wei-Meng Lee WILEY India Edition WROX Publication
2. Professional Android 4 Application Development, By Reto Meier WROX Publication
3. The official site for Android developers - https://developer.android.com

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Sem-VI (CBCS 2019 Pattern)
Subject Code: CA-604
Subject: Dot Net Framework

Total Hours: 48 **Total Credits: 3**

Course Prerequisites:

Student should have basic knowledge of:

- Visual Basic
- HTML
- Object Oriented concepts
- Ms-Access, Mysql, SQL Server

Course Objectives:

- To learn Microsoft framework architecture.
- Understand development of windows application.
- To learn data access mechanism.
- Create and consume libraries.
- Create a web application.
- To develop the website and application.

Course Outcome:

- Use the features of Dot Net Framework along with the features of VB, C# and ASP
- Design and develop window based and web based .NET applications.
- Design and develop a Website.
- Design and Implement database connectivity using ADO.NET for VB, C# and ASP.

Sr.No	Chapter Name	No.of Lectures
1	**Introduction to DOT NET FRAMEWORK** 1.1 What is Framework? 1.2 Architecture of Dot Net Framework 1.2.1 Common Language Runtime 1.2.2 Common Type System(CTS) 1.2.3 Common Language Specification(CLS) 1.2.3 JIT Compilers 1.2.3 Base Class Library 1.3 IDE (Integrated Development Environment) 1.4 Event Driven Programming	5
2	**Introduction to VB.Net** 2.1 Basics of VB.Net 2.1.1 Operators 2.1.2 Data Types 2.1.3 Control Structures 2.2Build Windows Applications 2.2.1 Controls: Form, TextBox, Button, Label, CheckBox, ListBox, ComboBox, RadioButton, DateTimePicker, MonthCalender, Timer, Progressbar,Scrollbar, PictureBox, ImageBox, ImageList, TreeView, ListView, Toolbar, StatusBar, Datagridview 2.2.2 Menus and PopUp Menu 2.2.3 Predefined Dialog controls: Color,Save,File,Open, Font 2.2.4 DialogBox - InputBox(), MessageBox, MsgBox()	11
3	**Introduction to C#** 3.1 Language Fundamentals 3.1.1 Data type and Control Constructs 3.1.2 Value and Reference Types,Boxing 3.1.3 Arrays 3.1.4 String class and its various operations 3.1.5 Functions 3.2 Object Oriented Concepts 3.2.1 Defining classes and Objects	12

	3.2.2 Access modifiers 3.2.3 Constructors 3.2.4 Inheritance 3.2.5 Interface 3.2.6 Abstract Class 3.2.7 Method Overloading and Overriding 3.2.8 Delegates	
4	**Introduction to ASP.NET** 4.1 What isASP.NET? 4.2 ASP.NET Page Life Cycle 4.3 Architecture ofASP.NET 4.4 Forms, WebPages, HTML forms, 4.5 Request & Response in Non-ASP.NET pages 4.6 Using ASP.NET Server Controls 4.7 Overview of Control structures 4.8 Functions 4.9 HTML events 4.9.1 ASP.NET Web control events 4.9.2 Event driven programming and postback 4.10 Introduction to Web forms 4.10.1 Web Controls 4.10.2 Server Controls 4.10.3 Client Controls 4.10.4 Navigation Controls 4.10.5 Validations 4.10.6 Master Page 4.10.7 State Management Techniques	10
5	**Architecture of Ado.Net** 5.1 Basics of Ado.net 5.1.1Connection Object 5.1.2Command Object 5.1.3Dataset 5.1.4Data Table 5.1.5Data Reader Object 5.1.6Data Adapter Object 5.2 Datagridview& Data Binding: Insert, Update, Delete records 5.3 Navigation Using Data Source	10
Total		48

Reference Books:

- Beginning Visual C#, WroxPublication
- **BeginningASP.NET3.5,WroxPublication**
- **ProgrammingASP.NET3.5byJesseLiberty,DanMaharry,DanHurwitz,O'Reilly**
- Programming Microsoft Visual Basic.NET – Francesco Balena
- The Complete Reference -Visual Basic .NET – Jefrey R. Shapiro
- ADO.NET Examples and Best Practices for C# Programmers, By Peter D, Blackburn, William
- VB.NET database programming with ADO.NET -Anne Prince and Doug Lowe

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Semester-VI
Subject: Project
Course Code : DSE– 605
Total Credits: 04

For the evaluation / conduction of project separate guidelines will be provided.

T.Y.B.B.A.(C.A.) Semester-VI
Subject: Computer Laboratory Based on 603 and 604(2 credits each)
Course Code: 606
Total Credits: 04

For the conduction of practical's, Practical Assignments are given in the Lab book.

Savitribai Phule Pune University
T.Y.B.B.A.(C.A.) Semester-VI
Subject: Soft Skill
Course Code : CA – 607

Total Hours: 30 **Credit:02**

Prerequisite:

1. Basic Writing Skills in English including grammar.
2. Basic knowledge in communication and a good understanding of English.
3. Ready to adhere the new techniques.

Objectives:

1. It helps participants to communicate effectively and to carry themselves confidently.
2. They also learn how to identify and overcome the barriers in interpersonal relationships.

3. To improve oral and written communication, teamwork, leadership, problem-solving and decision-making skills, to gain best results.
4. This course is useful for landing a great job, building a career and also finding employment as soft skills trainers.

Outcomes:

1. Understand the significance and essence of a wide range of soft skills
2. Learn how to apply soft skills in a wide range of routine social and professional settings.
3. Learn how to employ soft skills to improve interpersonal relationships.
4. Learn how to employ soft skills to enhance employability and ensure workplace and career success.

Unit	Topics	No. of Lectures
1	**Introduction to Soft Skills** 1.1 An Introduction to Soft skill - 1.1.1 Definition and Significance of Soft Skills 1.1.2 Soft skill Process 1.1.3 Uses of Soft Skill Development.	02
2	**Communication Skills** 2.1 Introduction - Components of communication process, Communication process , Effective communication process. 2.2 Types of communication – 2.2.1 Verbal Communication – • Punctuation • Meaning & opposites , vocabulary • Real Life conversations 2.2.2 Non – Verbal Communication - • Facial Expression , Posture , Gesture , Eye contact • appearance (dress code), Body Language, listening skills • essential formal writing skills	04
3	**Skills Development** 3.1 Interview Skills – Interviewer and Interviewee – in-depth perspectives. Before, During and After the Interview. Tips for Success. 3.2 Presentation Skills - Types, Content, Audience Analysis, Essential Tips Before, During and After, Overcoming Nervousness. 3.3 Etiquette and Manners - Social and Business 3.4 Time Management - Concept, Essentials, Tips 3.5 Personality Development - Meaning, Nature, Features,	05

	Stages, Models, Learning Skills, Adaptability Skills.	
4	**Skill Implementation** 4.1 Resume writing – 4.1.1 How to write your resume. • Contact details. • Opening statement. • List of key skills. • List of technical/software skills. • Personal attributes/career overview. • Educational qualifications. • Employment history /volunteering/work placements. • References/referees. 4.1.2 Types of resume 4.2 Group Discussion - Importance, Planning, Elements, and Skills assessed, Effectively disagreeing, Initiating, Summarizing and Attaining the Objective. 4.3 Teamwork and Leadership Skills - Concept of Teams, Building effective teams, Concept of Leadership and honing Leadership skills , A Good Leader, Leaders and Managers , Types of Leaders , Leadership Behaviour.	04
Total		15
Practical Please Refer Lab Book		15

Reference Books :

1. Managing Soft Skills for Personality Development – edited by B.N.Ghosh, McGraw Hill India, 2012.
2. English and Soft Skills – S.P.Dhanavel, Orient Blackswan India, 2010.
3. Soft skills Training – A workbook to develop skills for employment by Fredrick H. Wentz .
4. Personality Development and Soft skills, Oxford University Press by Barun K. Mitra
5. The Time Trap : the Classic book on Time Management by R. Alec Mackenzie